



KTU NOTES

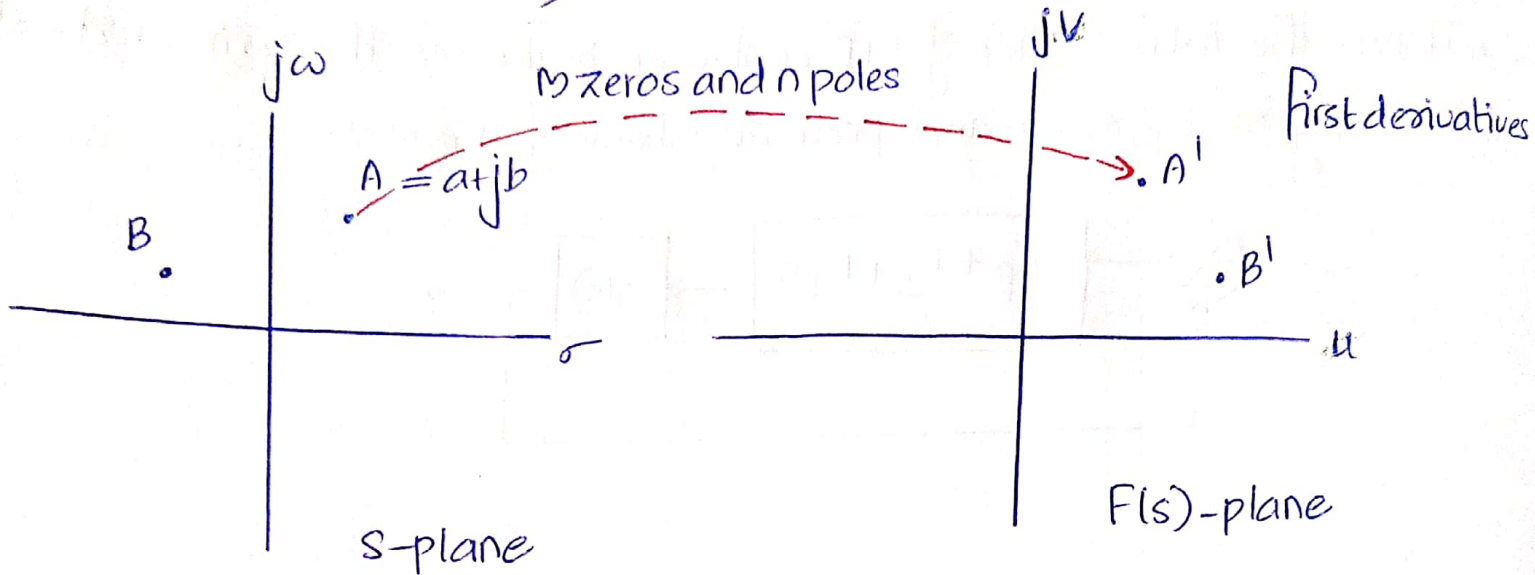
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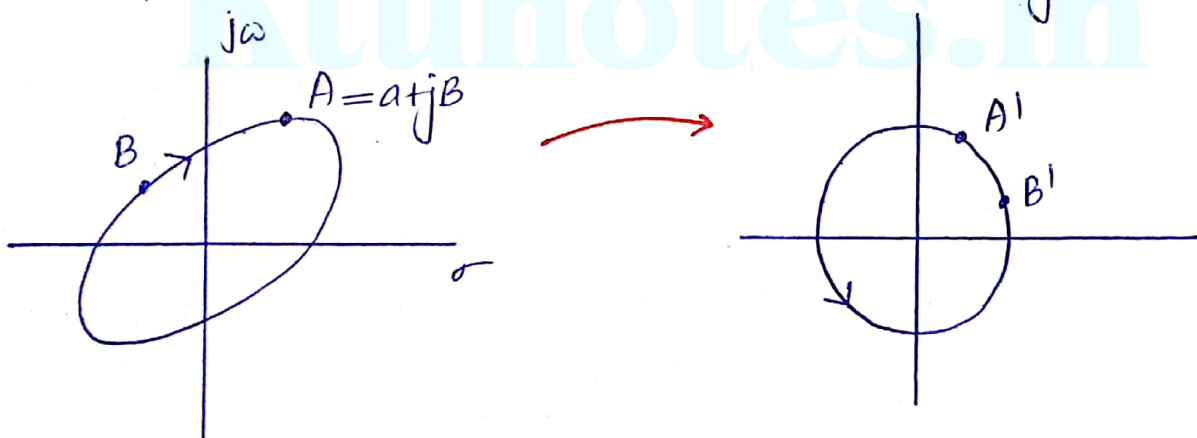
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MODULE-4 NYQUIST STABILITY CRITERION

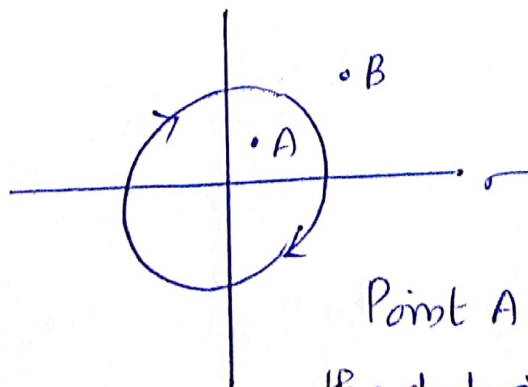
$$F(s) = \frac{(s-z_1)(s-z_2)\dots(s-z_m)}{(s-p_1)(s-p_2)\dots(s-p_n)} = u+jv$$



→ For all points in s plane, there is a corresponding mapping in $F(s)$ plane.

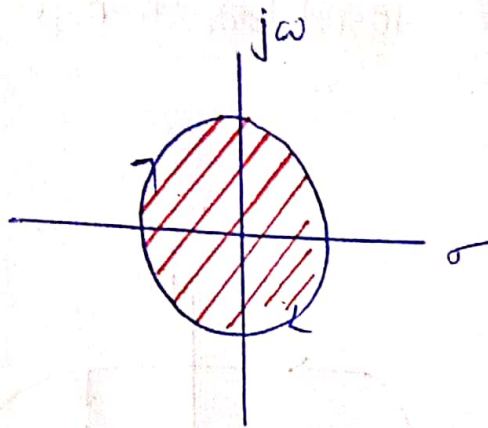


→ **Encirclement:** A point is said to be encircled by a closed path if it is found inside the path.

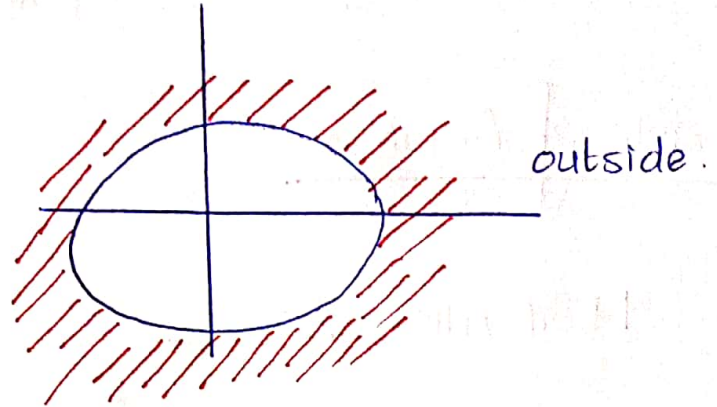


Point A is encircled by the contour in the clockwise direction. B - not encircled.

→ **Enclosurement**: A point is said to be enclosed by a closed path, if it is found to lie to the right of the path when the path is traversed in the prescribed direction.



All points inside are enclosed



Development of Nyquist Stability Criterion

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

characteristic eqn: $1 + G(s) \cdot H(s) = 0$

$$\text{let } G(s) \cdot H(s) = \frac{k(s + z_1')(s + z_2') \dots (s + z_m')}{(s + p_1)(s + p_2) \dots (s + p_n)}$$

$$\begin{aligned} 1 + G(s) \cdot H(s) &= \frac{1 + k(s + z_1')(s + z_2') \dots (s + z_m')}{(s + p_1)(s + p_2) \dots (s + p_n)} \\ &= \frac{(s + p_1)(s + p_2) \dots (s + p_n) + k(s + z_1')(s + z_2') \dots (s + z_m')}{(s + p_1)(s + p_2) \dots (s + p_n)} \\ &= \frac{(s + z_1)(s + z_2) \dots (s + z_m)}{(s + p_1)(s + p_2) \dots (s + p_n)} \end{aligned}$$

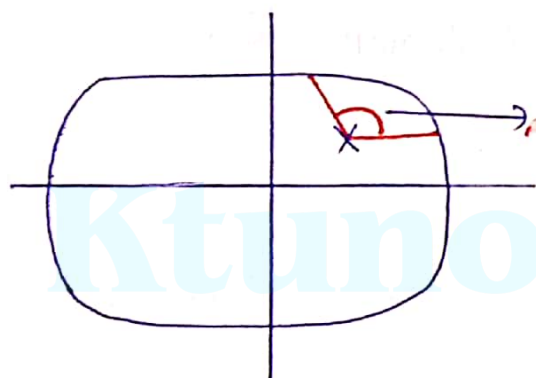
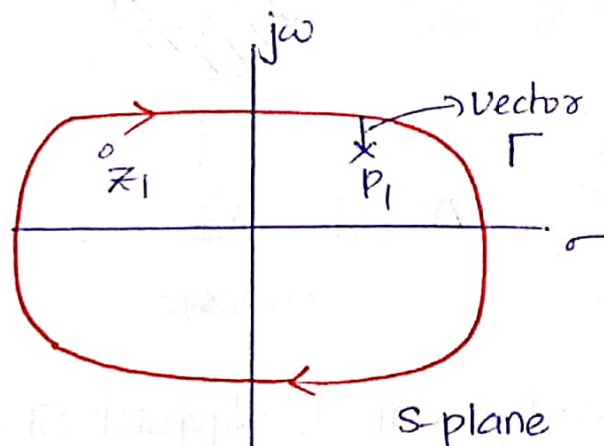


Poles of $1 + G(s) \cdot H(s) \equiv$ Poles of $G(s) \cdot H(s)$
 zeros of $1 + G(s) \cdot H(s) \equiv$ Roots of the characteristic eqn $\equiv$
 poles of closed loop Transfer Function

Principle of Argument

$$1 + G(s)H(s) = Q(s) = \frac{s + z_1}{s + p_1}$$

Zero, z_1 and pole, p_1 is enclosed by the contour Γ .



All angles are +ve in anti-clockwise direction.

→ When the vector moves through the closed path, net change in angle $\rightarrow 360^\circ$ rotation.

Vector makes 2π angle in clockwise direction.

$$\delta_{\Gamma} \angle s + p_1 = -2\pi$$

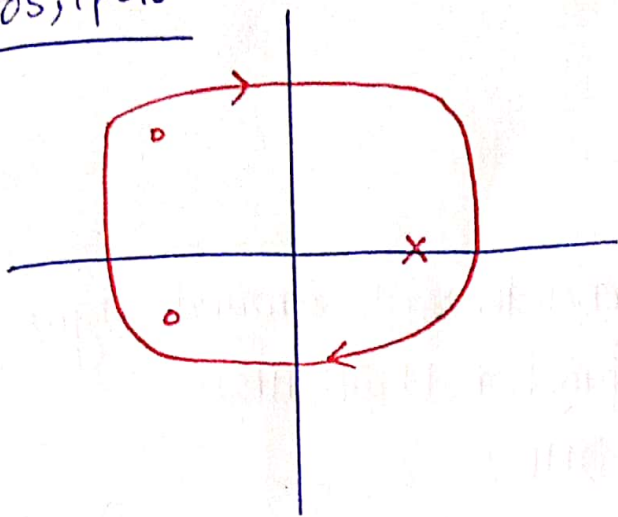
$$\delta_{\Gamma} \angle s + z_1 = -2\pi$$

→ -ve $\Rightarrow$ clockwise

+ve $\Rightarrow$ anti-clockwise

$$\begin{aligned} \delta_{\Gamma} \angle Q(s) &= \delta_{\Gamma} (s + z_1) - \delta_{\Gamma} (s + p_1) \\ &= -2\pi - (-2\pi) \\ &= \underline{\underline{0}} \end{aligned}$$

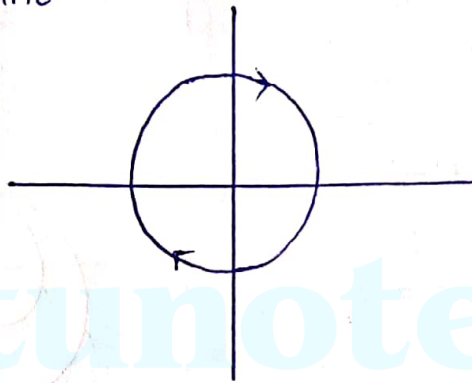
2 zeros, 1 pole



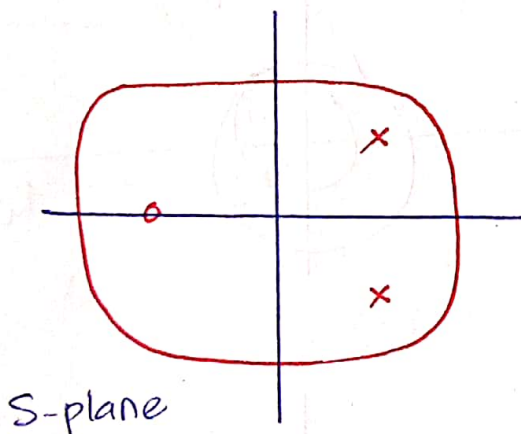
$$\begin{aligned}\delta_f \angle Q(s) &= \delta_f \angle s + z_1 + \delta_f \angle s + z_2 \\ &\quad - \delta_f \angle s + p_1 \\ &= -2\pi\end{aligned}$$

→ If we map points from s-plane to $Q(s)$ plane:

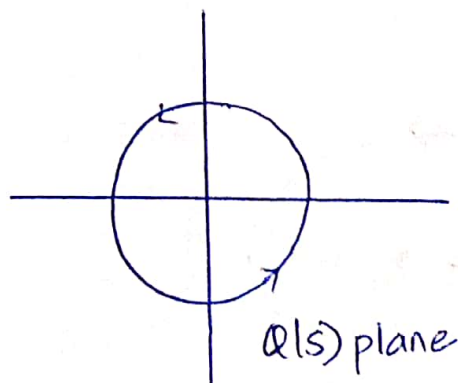
$-2\pi \Rightarrow$ one clockwise direction about origin in the $Q(s)$ plane.



2 poles, 1 zero



$$\begin{aligned}\delta_f \angle Q(s) &= -2\pi - (-4\pi) \\ &= 2\pi\end{aligned}$$



Anti-clockwise rotation about origin

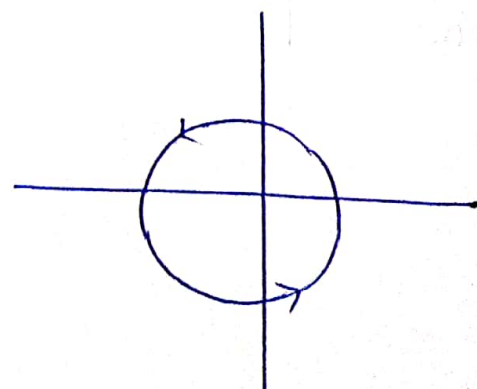
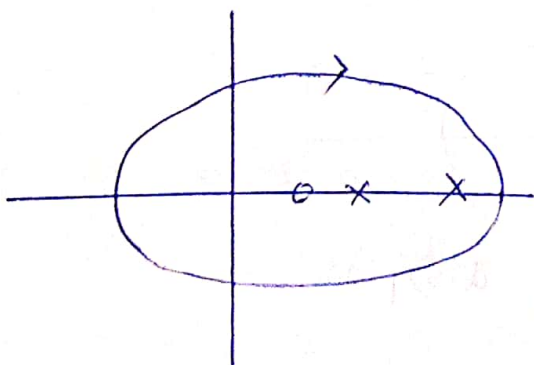
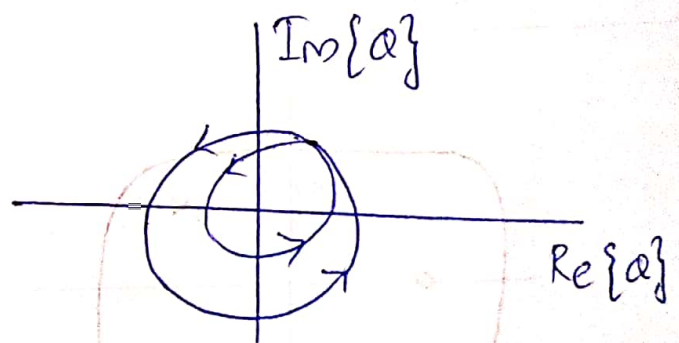
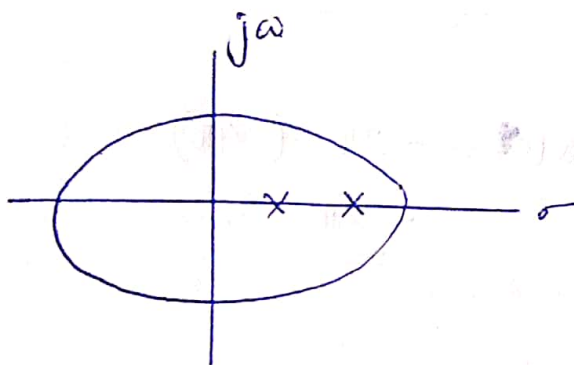
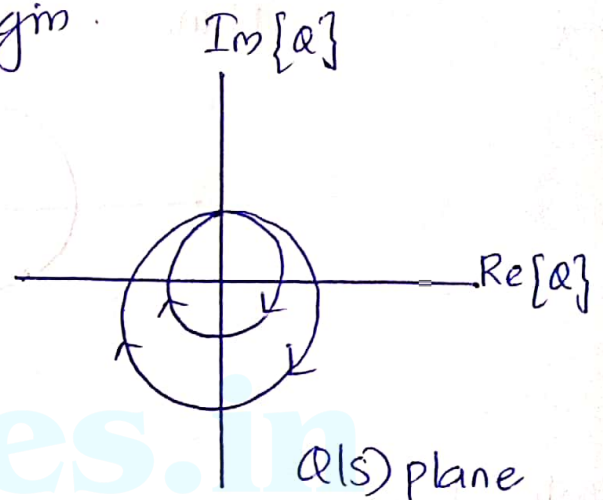
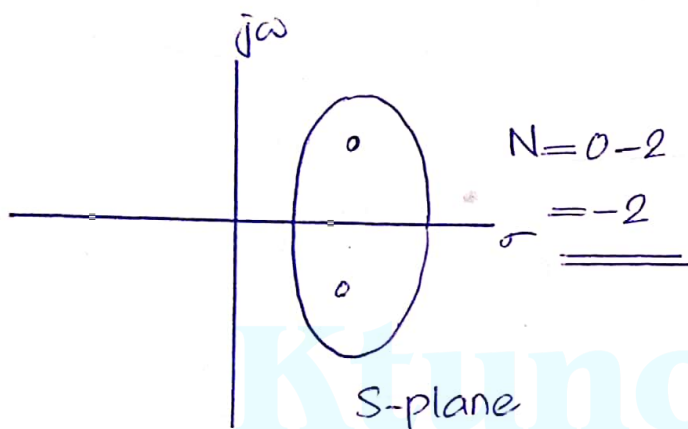
Generally,

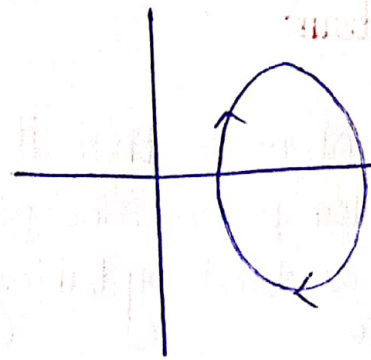
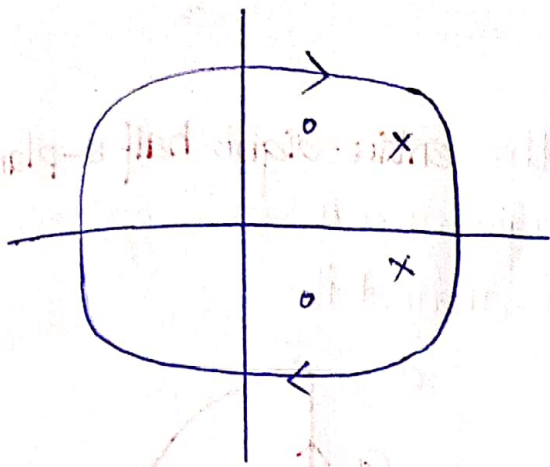
$$N = P - Z \text{ where}$$

N is the no. of anticlockwise encirclements around origin
 P is the no. of poles of the function $1 + G(s)H(s)$.
 Z is the no. of zeros of $1 + G(s)H(s)$

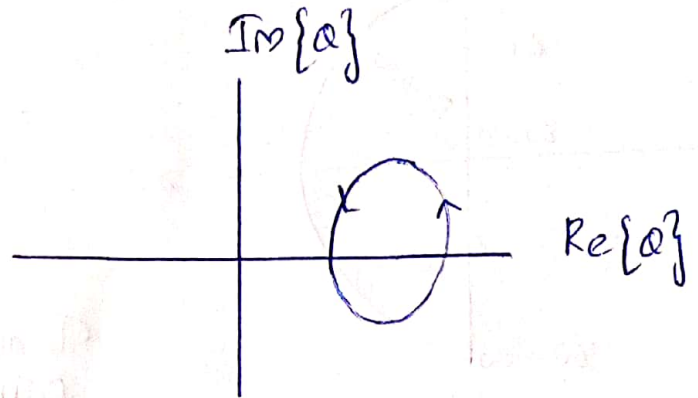
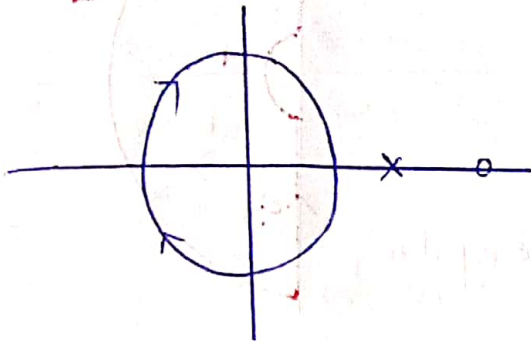
- 2 poles, 1 zero: $N = 2 - 1 = 1$ So, 1 anticlockwise encirclement.
- 1 pole, 2 zeros: $N = -1$ clockwise encirclement

$N = 0 \Rightarrow$ No encirclement around origin.





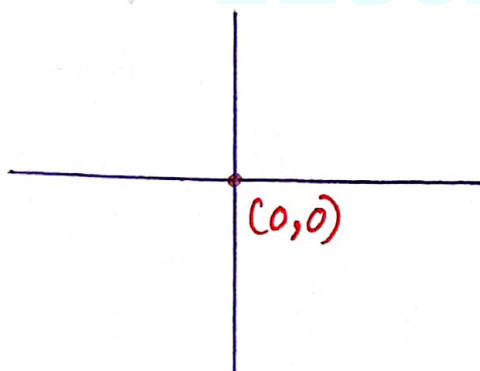
No encirclement around origin



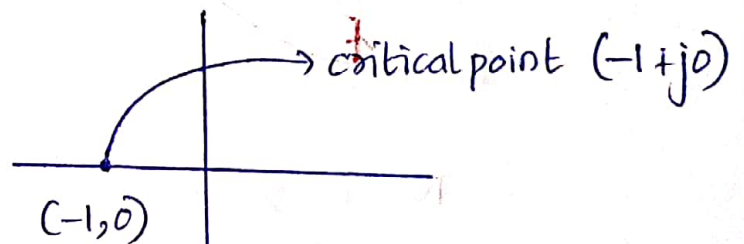
$1 + G(s) \cdot H(s) = Q(s) \rightarrow$ encirclements around origin

OLTF $G(s)H(s) = Q(s) - 1 \rightarrow$ point will be shifted 1 unit to the left.

$G(s)H(s)$ plane



$Q(s)$ plane

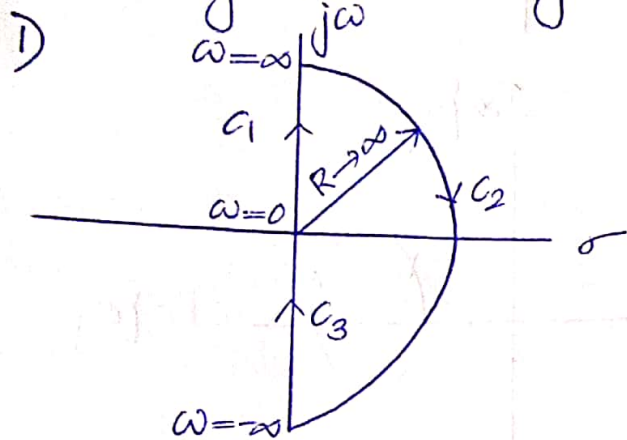


$Q(s)-1$ plane

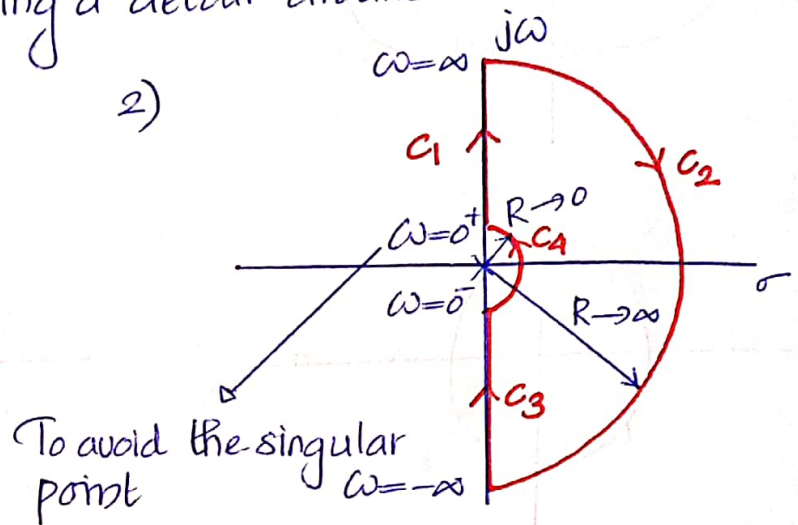
We look for no. of encirclements around critical point if we refer to $G(s)H(s)$ function.

Nyquist Contour

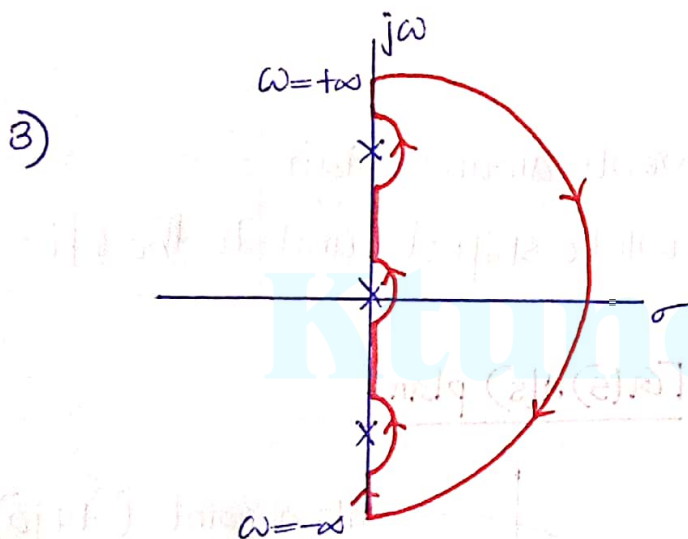
Select nyquist contour so that it encloses the **entire right half s-plane**, except the singular points. The poles on imaginary axis are singular points and so they are avoided by taking a detour around it.



2)



$$G(s) \cdot H(s) = \frac{1}{s}$$



Singular points should be avoided

$$N = P - Z$$

N is the no. of anti-clockwise encirclements

P is the no. of poles of $1 + G(s)H(s)$ on right half of s-plane

$\equiv$ Poles of $G(s) \cdot H(s)$

Z is the no. of zeros of $1 + G(s)H(s)$ on right half of s-plane

$\equiv$ poles of CLTF on right half of s-plane

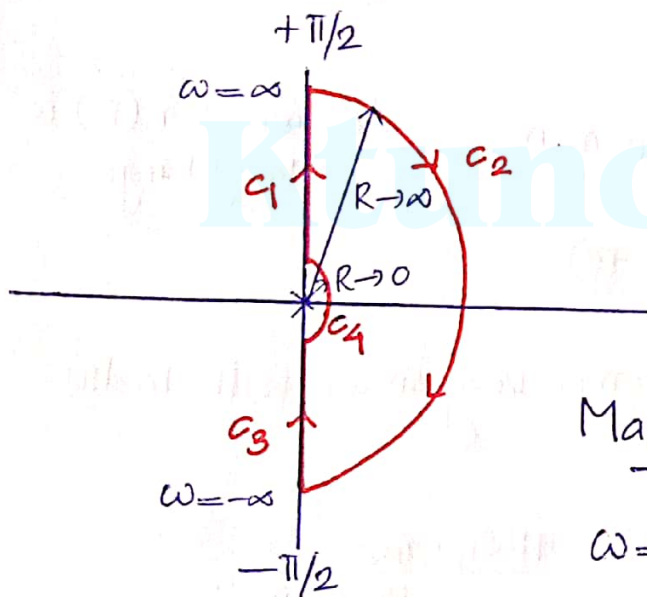
$\equiv$ roots of characteristic eqn.

• For a stable system, $Z = 0$ $N = P$

Nyquist Stability Criterion

- 1) If there are no encirclements around $-1+j0$
 $N=0, P=0$. If there are no poles at $G(s)H(s)$ on r.h of s plane.
- 2) If there are anti-clockwise encirclements, s/m becomes stable when $N=P$.
- 3) Clockwise encirclements around $-1+j0$.
Indicates the presence of zeros
 $N=-1$ {For eg}
 s/m is unstable always.

Procedure



Mapping of C_1 into $G(s)H(s)$ plane

$\omega = 0^+$ to $\omega = +\infty$

No real parts. so, substitute $s=j\omega$ in $G(s)H(s)$

$$\Rightarrow G(j\omega)H(j\omega)$$

Then for various values of ω , sketch $G(j\omega)H(j\omega)$

Mapping of C_3 into $G(s)H(s)$ plane

$\omega = -\infty$ to $\omega = 0^-$

Mirror image of C_1 w.r.t real axis.

Mapping of C_2

$$\text{Let } G(s) \cdot H(s) = \frac{k(1+ST_1)(1+ST_2) \dots (1+ST_m)}{s^p(1+ST_a)(1+ST_b) \dots (1+ST_n)}$$

m - zeros
 n - poles

$$s = \lim_{R \rightarrow \infty} R \cdot e^{j\theta}$$

θ varies from $+\frac{\pi}{2}$ to $-\frac{\pi}{2}$.

Substitute $s = \lim_{R \rightarrow \infty} R \cdot e^{j\theta}$ in $G(s)H(s)$

$\rightarrow 1+ST \approx ST$ because R value is very high

$$G(s) \cdot H(s) \approx \frac{kST_1 \cdot ST_2 \dots ST_m}{s^p \cdot ST_a \cdot ST_b \dots ST_n} = \frac{k_1 s^m}{s^n} = \frac{k_1}{s^{n-m}}$$

$$G(s) \cdot H(s) \Big|_{s = \lim_{R \rightarrow \infty} R e^{j\theta}} = \frac{k_1}{\lim_{R \rightarrow \infty} (R \cdot e^{j\theta})^{n-m}} = 0 \cdot e^{-j\theta(n-m)}$$

denominator (R) is very large.

In the $G(s)H(s)$ plane, the arc becomes very small with radius tending to 0.

S-plane

Starting pt: When $\theta = +\frac{\pi}{2}$
ending pt: When $\theta = -\frac{\pi}{2}$

$G(s)H(s)$ plane

$$G(s)H(s) = 0 \cdot e^{-j\frac{\pi}{2}(n-m)}$$
$$G(s)H(s) = 0 \cdot e^{+j\frac{\pi}{2}(n-m)}$$

Mapping of C_4

$$G(s)H(s) = \frac{k(1+ST_1)(1+ST_2) \dots}{s^p(1+ST_a)(1+ST_b)}$$

m - zeros
 n - poles

$$S = \lim_{R \rightarrow 0} R e^{j\theta}$$

$$\theta = -\frac{\pi}{2} \text{ to } \frac{\pi}{2}$$

$$HST \approx 1$$

$$G(s)H(s) = \frac{k}{s^p}$$

$$S = \lim_{R \rightarrow 0} R e^{j\theta}$$

$$G(s)H(s) = \frac{k}{\lim_{R \rightarrow 0} (R e^{j\theta})^p} = \frac{k}{\lim_{R \rightarrow 0} R^p e^{j\theta p}} = \frac{k}{0} = \infty \cdot e^{-j\theta p}$$

In $G(s)H(s)$ plane, the arc becomes very large with $R \rightarrow \infty$

Starting point

s-plane

$$\text{When } \theta = -\frac{\pi}{2},$$

$G(s)H(s)$ plane

$$G(s)H(s) = \infty \cdot e^{j\frac{\pi}{2}p}$$

Ending point

s-plane

$$\text{When } \theta = \frac{\pi}{2}$$

$G(s)H(s)$ plane

$$G(s)H(s) = \infty \cdot e^{-j\frac{\pi}{2}p}$$

→ Arc — radius ∞

$$+\frac{\pi}{2}p \text{ to } -\frac{\pi}{2}$$

p is the no. of poles at origin

Qn) Draw the nyquist plot for the s/m whose OLTF is

$$G(s)H(s) = \frac{k}{s(s+2)(s+10)}$$

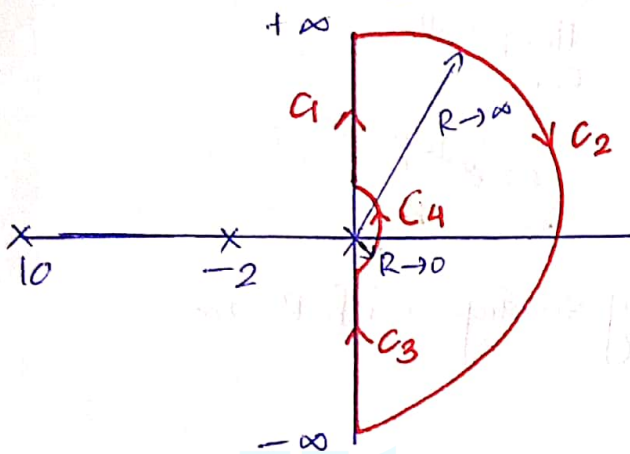
Determine the range of k for which the closed loop s/m is stable.

Step 1: Convert OUTF to standard form

$$G(s)H(s) = \frac{k}{s \times 2(1+0.5s) \times 10(1+0.1s)}$$

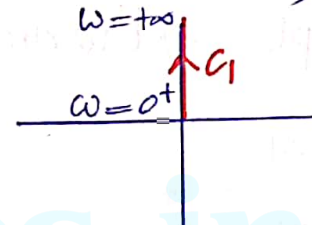
$$= \frac{0.05k}{s(1+0.5s)(1+0.1s)}$$

Step 2: Select nyquist contour



Mapping of C_1

$$G(s)H(s) = \frac{0.05k}{s(1+0.5s)(1+0.1s)}$$



$$s = j\omega;$$

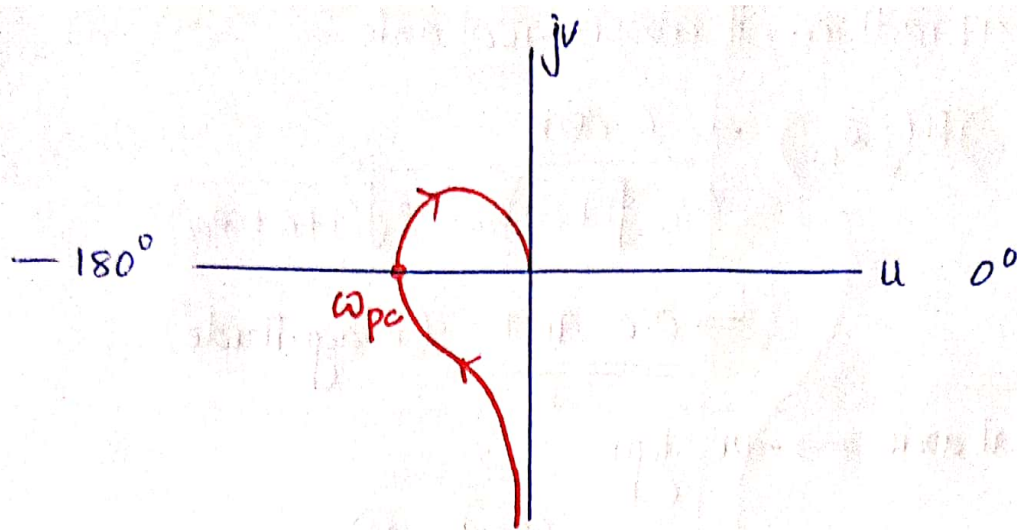
$$G(j\omega)H(j\omega) = \frac{0.05k}{j\omega(1+0.5j\omega)(1+0.1j\omega)}$$

$$= \frac{0.05k}{\omega \sqrt{1+(0.5\omega)^2} \sqrt{1+(0.1\omega)^2}}$$

$$\angle G(j\omega)H(j\omega) = 0^\circ - 90^\circ - \tan^{-1}\left(\frac{0.5\omega}{1}\right) - \tan^{-1}\left(\frac{0.1\omega}{1}\right)$$

$$= -90^\circ - \tan^{-1}(0.5\omega) - \tan^{-1}(0.1\omega)$$

ω	$ G(j\omega)H(j\omega) $	$\angle G(j\omega)H(j\omega)$
$\omega = 0$	∞	-90°
$\omega = 1$	$0.0445k$	-122°
$\omega = 5$	$0.003k$	-185°
$\omega = \infty$	0	-270°



Phase Cross over Frequency: Frequency at which phase of s/m crosses $\pm 180^\circ$ (ω_{pc})

$$\begin{aligned}
 G(j\omega)H(j\omega) &= \frac{0.05k}{(j\omega + (-1)\omega^2 0.5)(1 + 0.1j\omega)} \\
 &= \frac{0.05k}{(j\omega - 0.5\omega^2)(1 + 0.1j\omega)} \\
 &= \frac{0.05k}{j\omega - 0.1\omega^2 - 0.5\omega^2 - 0.05j\omega^2} \\
 &= \frac{0.05k}{-0.6\omega^2 + j\omega(1 - 0.05\omega^2)}
 \end{aligned}$$

When $\omega = \omega_{pc}$, $\omega(1 - 0.05\omega^2) = 0$

$$\omega_{pc}(1 - 0.05\omega_{pc}^2) = 0$$

$$1 - 0.05\omega_{pc}^2 = 0$$

$$0.05\omega_{pc}^2 = 1$$

$$\omega_{pc}^2 = \frac{1}{0.05} = 20$$

$$\omega_{pc} = 4.47 \text{ rad/sec}$$

At $\omega_{pc} = 4.47 \text{ rad/sec}$, it crosses real axis

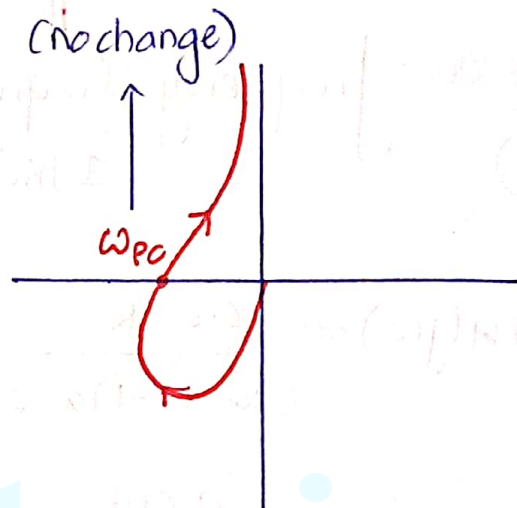
$$G(j\omega_{pc})H(j\omega_{pc}) = \frac{0.05k}{\omega_{pc}\sqrt{1+(0.5\omega_{pc})^2}\sqrt{1+0.1\omega_{pc}^2}}$$

$$= \underline{\underline{0.00477k}} \text{ (Magnitude)}$$

-ve real axis $\Rightarrow$ -ve sign

Mapping of C_3

Mirror image
 $\omega = -\infty$ to $\omega = 0^-$



Mapping of C_2

$$S = \lim_{R \rightarrow \infty} R e^{j\theta}$$

$$\theta = +\frac{\pi}{2} \text{ to } -\frac{\pi}{2}$$

$$G(s) \cdot H(s) = \frac{k}{s(s+2)(s+10)} = \frac{k}{s \times 2(1+0.5s)10(1+0.1s)}$$

$$= \frac{0.05k}{s(1+0.5s)(1+0.1s)}$$

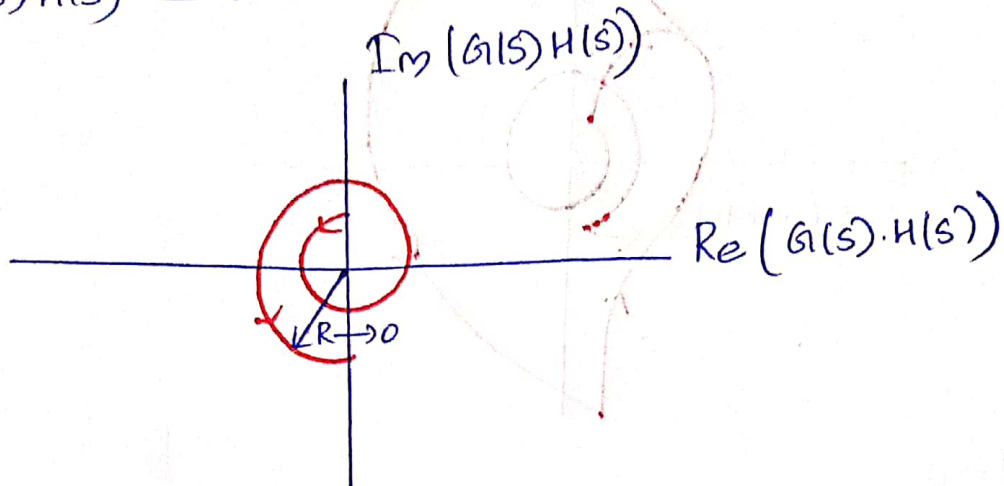
$$R \gg 1 \Rightarrow 1+ST \approx ST$$

$$G(s) \cdot H(s) = \frac{0.05k}{s \times 0.5s \times 0.1s} = \frac{k}{s^3}$$

$$= \frac{k}{\lim_{R \rightarrow \infty} (R e^{j\theta})^3} = \frac{k}{\lim_{R \rightarrow \infty} R^3 e^{j3\theta}} = 0 \cdot e^{-j3\theta}$$

$$\theta = \frac{\pi}{2}, G(s)H(s) = 0 \cdot e^{-j3\pi/2}$$

$$\theta = -\frac{\pi}{2}, G(s)H(s) = 0 \cdot e^{j3\pi/2}$$



Mapping of C_4

$$s = \lim_{R \rightarrow 0} R e^{j\theta}$$

$$\theta = -\frac{\pi}{2} \text{ to } \frac{\pi}{2}$$

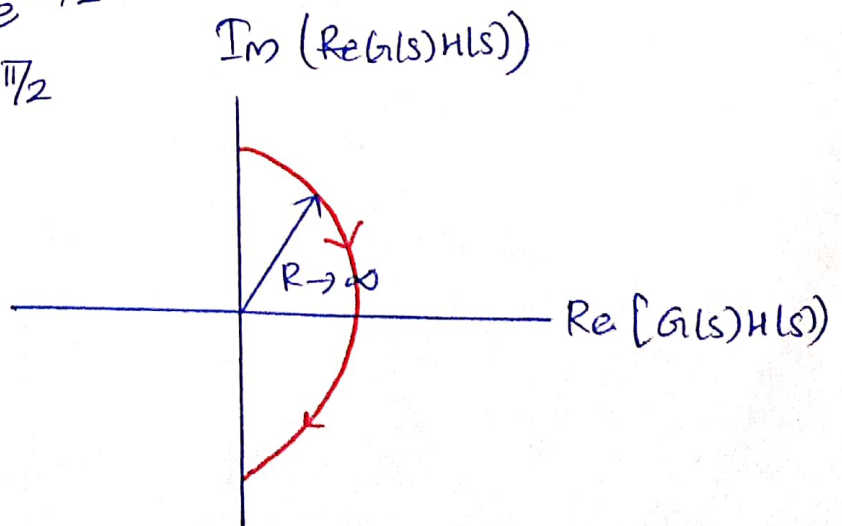
$$1 + ST \approx 1$$

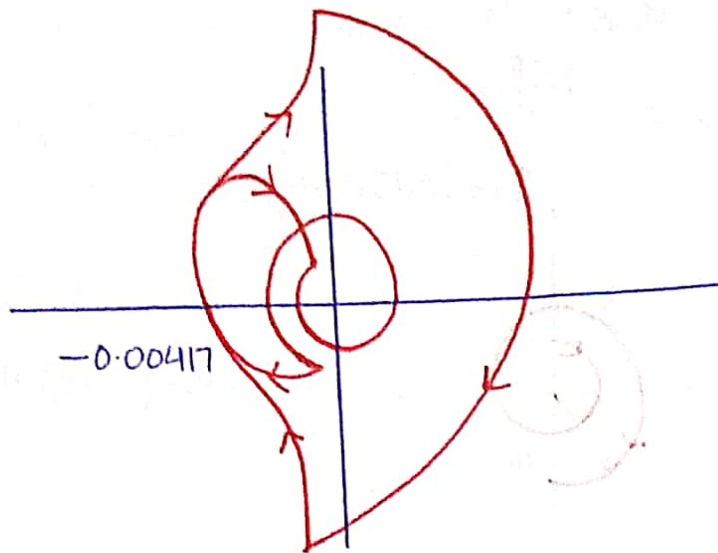
$$G(s)H(s) = \frac{0.05k}{s(1+0.5s)(1+0.1s)} = \frac{0.05k}{s}$$

$$G(s)H(s) = \frac{0.05k}{\lim_{R \rightarrow 0} R e^{j\theta}} = \infty \cdot e^{-j\theta}$$

$$\theta = -\frac{\pi}{2}, G(s)H(s) = \infty \cdot e^{j\pi/2}$$

$$\theta = \frac{\pi}{2}, G(s)H(s) = \infty \cdot e^{-j\pi/2}$$





$$P=0 \quad N=0$$

limiting value of k

$$-0.00417 k = -1$$

$$\underline{k = 240}$$

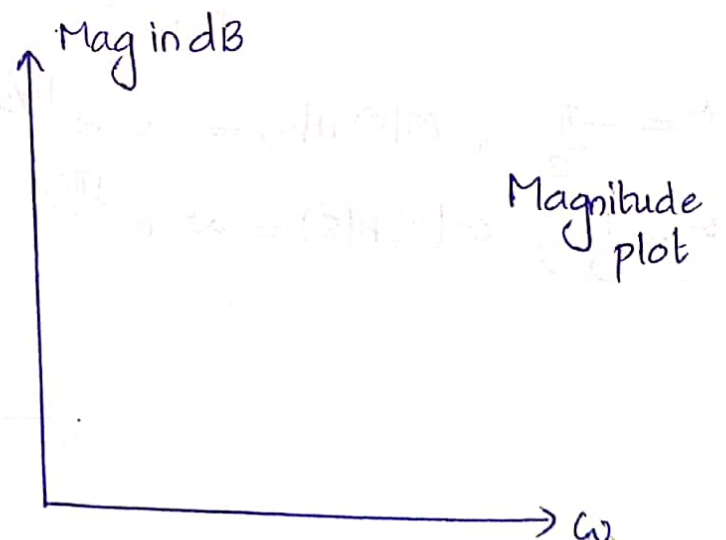
No encirclements around $(-1, 0)$

$$k < 240 \Rightarrow \begin{matrix} N=0 \\ P=0 \end{matrix} \left. \vphantom{\begin{matrix} N=0 \\ P=0 \end{matrix}} \right\} \text{stable s/m}$$

$k > 240 \Rightarrow 2$ clockwise encirclements
s/m always unstable.

Bode plots - Frequency response

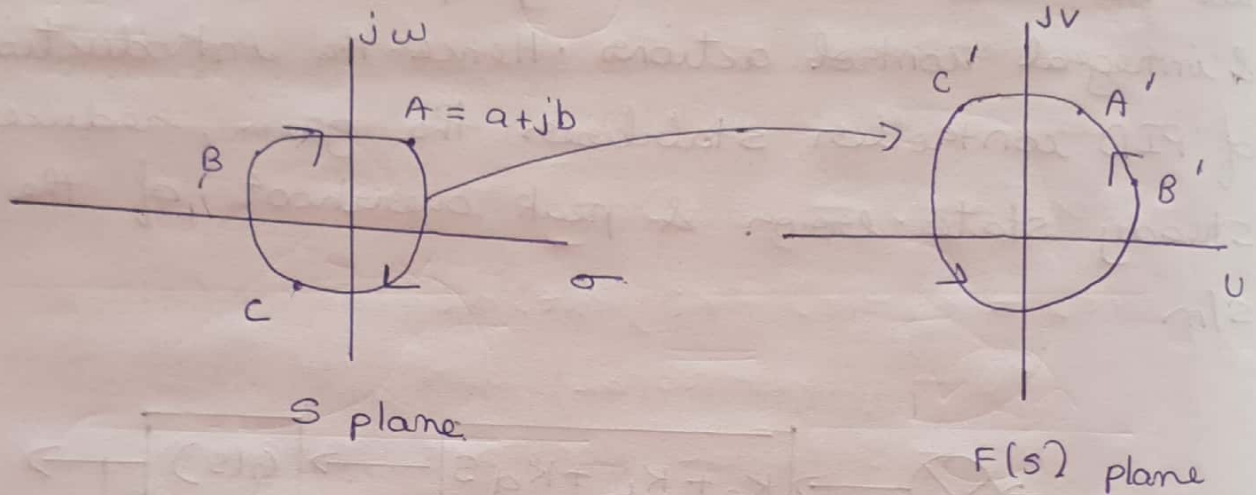
$$G(s)H(s) = \frac{k(1+sT_1)}{s(1+sT_2)}$$



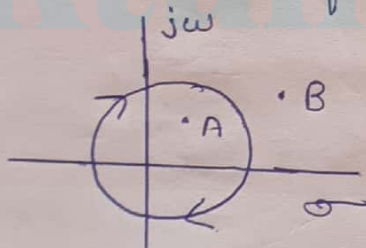
NYQUIST STABILITY CRITERION

$$F(s) = \frac{(s - z_1)(s - z_2) \dots (s - z_m)}{(s - p_1)(s - p_2) \dots (s - p_n)} = u + jv$$

m - zeros n - poles.



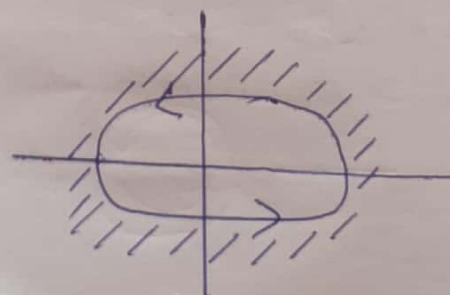
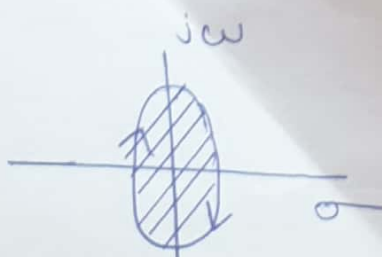
Encirclement - A point is said to be encircled by a closed path, if it is found inside the path.



A - encircled by the contour (path) in clock wise direction

B - Not encircled

Enclosurement - A point is said to be enclosed by a closed path, if it is found to lie to the right of the path when the path is traversed in the prescribed direction.



$$\frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s) \cdot H(s)}$$

characteristic eqn
 $= 1 + G(s) \cdot H(s) = 0$

Let $G(s) \cdot H(s) = \frac{K(s+z_1')(s+z_2') \dots (s+z_m')}{(s+p_1)(s+p_2) \dots (s+p_n)}$

$$1 + G(s)H(s) = 1 + \frac{K(s+z_1')(s+z_2') \dots (s+z_m')}{(s+p_1)(s+p_2) \dots (s+p_n)}$$

$$= \frac{(s+p_1)(s+p_2) \dots (s+p_n) + K(s+z_1')(s+z_2') \dots (s+z_m')}{(s+p_1)(s+p_2) \dots (s+p_n)}$$

$$= \frac{(s+z_1)(s+z_2) \dots (s+z_m)}{(s+p_1)(s+p_2) \dots (s+p_n)}$$

Poles of $1 + G(s) \cdot H(s)$ = Poles of $G(s) \cdot H(s)$

Zeros of $1 + G(s) \cdot H(s)$ = Roots of characteristic eqn.

$\equiv$ Poles of closed loop T-F

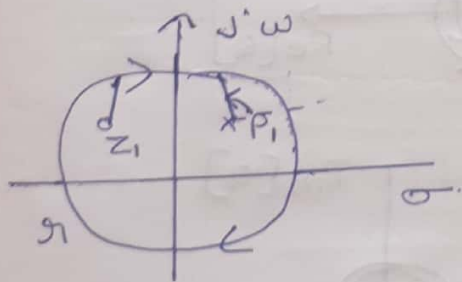
Principle of argument

$$N = P - Z$$

$$= 1 - 1$$

$$= 0$$

No encirclement.



Poles & zero are enclosed by the contour s

$$\oint_{\gamma} \frac{1}{s+p_1} = -2\pi$$

$$\oint_{\gamma} \frac{1}{s+z_1} = -2\pi$$

$$\oint_{\gamma} Q(s) = \oint_{\gamma} (s+z_1) - \oint_{\gamma} (s+p_1)$$

$$= -2\pi - (-2\pi) = 0 //$$

-ve sign
 $\downarrow$
 clockwise direction

$$\angle a(s) = \angle (s+Z_1) +$$

$$\angle (s+Z_2) - \angle (s+P_1)$$

$$= -2\pi + -2\pi - (-2\pi)$$

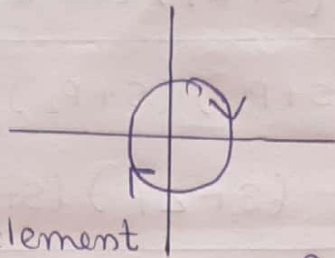
$$= \underline{\underline{-2\pi}}$$

$$N = P - Z$$

$$= 1 - 2$$

$$= \underline{\underline{-1}}$$

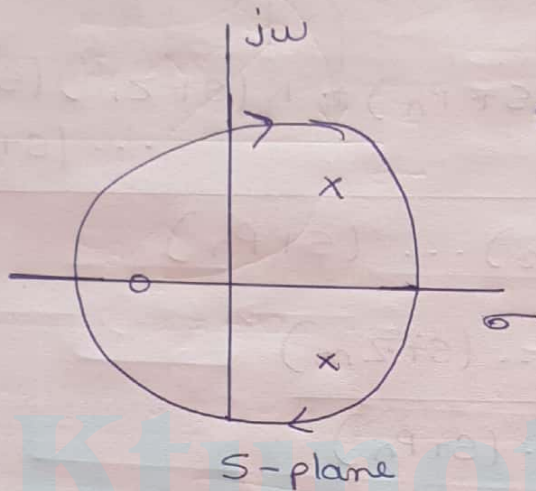
1 clockwise encirclement



clockwise

rotation.

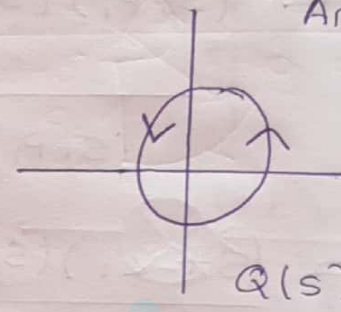
$Q(s)$ - plane



$$N = 2 - 1$$

$$= \underline{\underline{1}}$$

Anticlockwise



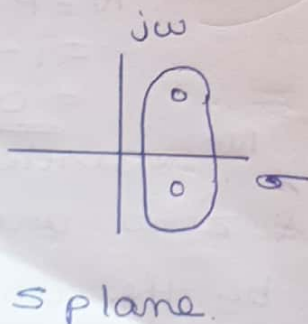
$Q(s)$ - plane

Ktunotes.in

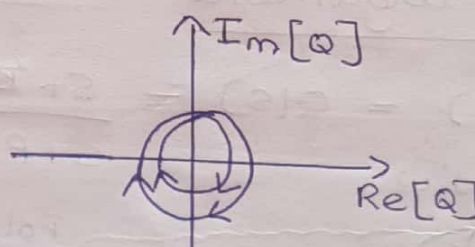
$$N = P - Z$$

N = no. of anticlockwise encirclement rotation around origin

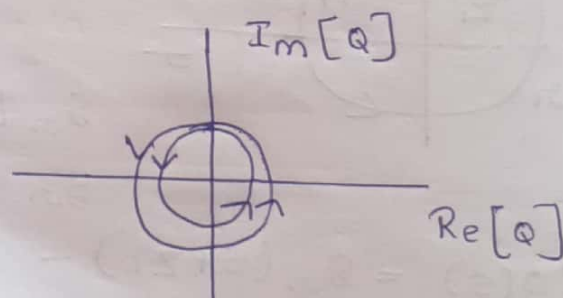
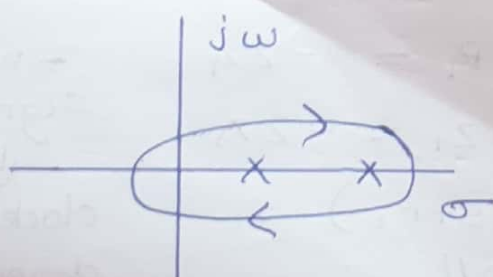
P = no. of poles of $1 + G(s) \cdot H(s)$

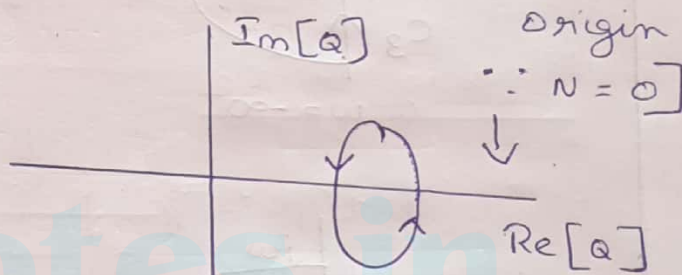
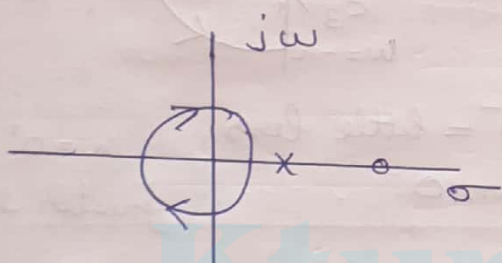
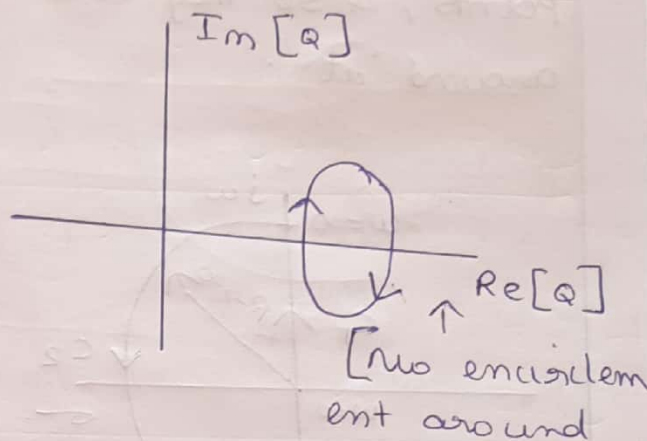
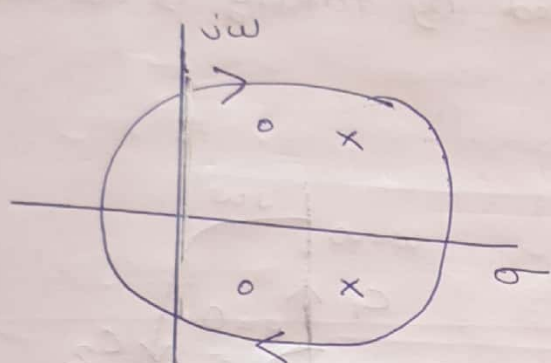
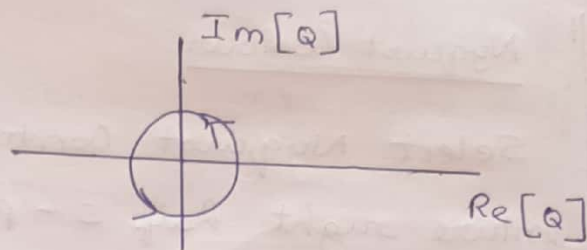
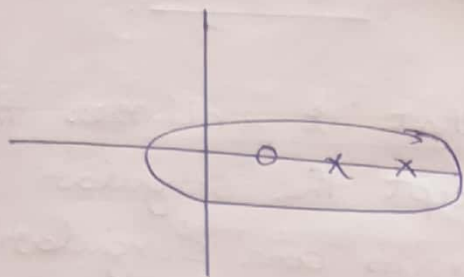


S plane.



$$N = -2$$





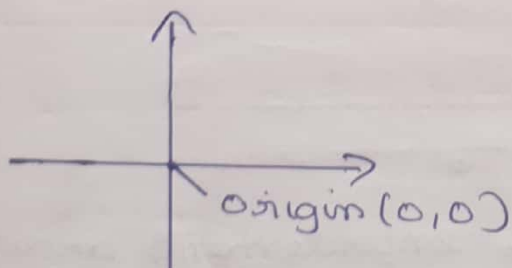
Stability based on O-L-T-F

$1 + G(s)H(s) = Q(s)$ — Encirclement around the origin

$$Q(s) - 1 = G(s) \cdot H(s)$$

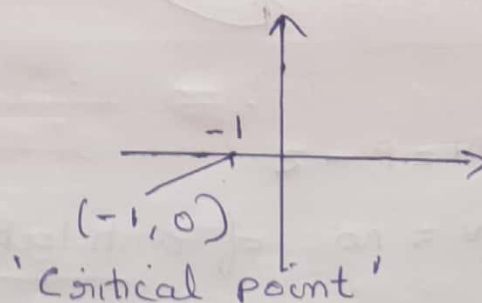
1

Encirclement around -1



$Q(s) - plane$

$-1 + j0$ — critical point

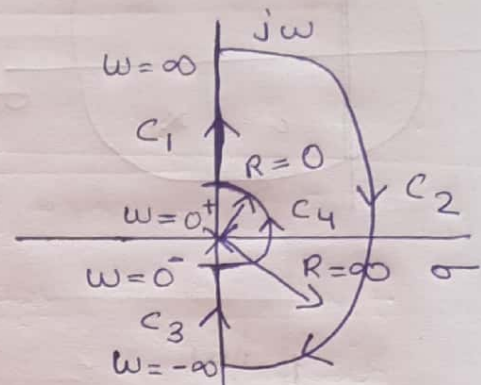
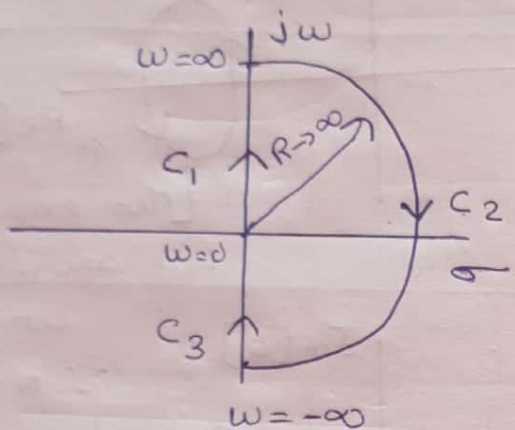


$Q(s) - 1 plane$

$G(s) \cdot H(s) plane$

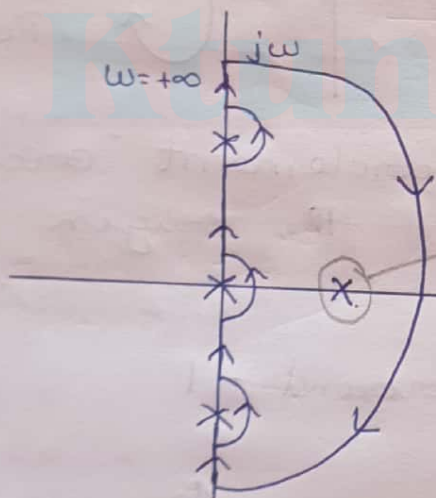
Nyquist Contour

Select Nyquist Contour so that it encloses the entire right half s -plane, except the singular points. The poles on imaginary axis are singular points, & so they are avoided by taking a detour around it.



$w=0^-$ = little less than 0

$w=0^+$ = little greater than 0



No need to avoid because not interrupting the path.
Only poles in imaginary axis should be avoided.

$$N = P - Z$$

N = no. of anticlockwise encirclements around critical

P - no. of poles of $1 + G(s)H(s)$ on right half of s -plane

$\equiv$ poles of $G(s)H(s)$ on right half of s -plane.

Z - no. of poles of $1 + G(s) \cdot H(s)$ on right half of s -plane $\equiv$ poles of C.LTF on right half of s -plane.

for stable S/m ; $Z = 0$

$$\boxed{N = P}$$

Nyquist Stability Criterion

If the $G(s)H(s)$ plane contour corresponding to Nyquist contour in the s -plane encircles the point $(-1+j0)$ in the anticlockwise direction as many times as the number of right half s -plane poles of O.L.T.F $G(s)H(s)$ then the closed loop S/m is stable.

For stability there should not be any zeros of $1 + G(s)H(s)$ on the right half of s -plane

$$N = P - Z$$

If $Z = 0$, then $N = P$

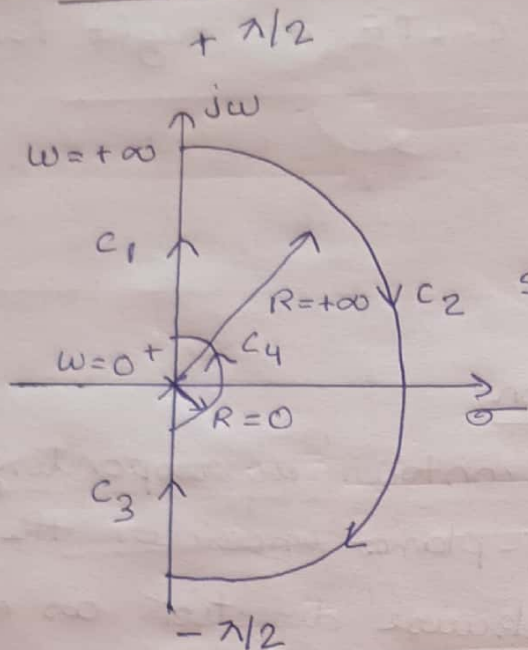
1) If there are no encirclement around $-1+j0$
 $N = 0$, $P = 0$

If there are no poles of $G(s)H(s)$ on right half of s -plane.

2) If there are anticlockwise encirclements
 $N = P$ — stable

3) clockwise encirclements around $-1+j0$
S/m always unstable.

Procedure



Mapping of C_1 into $G(s)H(s)$ plane

$$w = 0^+ \text{ to } w = +\infty$$

Sub $s = jw$ in $G(jw) \cdot H(jw)$ for various values of w sketch $G(jw) \cdot H(jw)$

Mapping of C_3 into $G(s)H(s)$ plane

$$w = -\infty \text{ to } 0^-$$

Mirror image of C_1 w.r.t real axis

Mapping of C_2

$$\text{Let } G(s)H(s) = \frac{K(1+ST_1)(1+ST_2)\dots(1+ST_m)}{s^p(1+ST_a)(1+ST_b)\dots(1+ST_n)}$$

m - Zeros
 n - poles

$$s = \lim_{R \rightarrow \infty} R \cdot e^{j\theta}$$

θ varies from $+\pi/2$ to $-\pi/2$

$\rightarrow$ Sub in $G(s)H(s)$

$$1+ST \approx ST$$

$$G(s) \cdot H(s) = \frac{K \cdot ST_1 \cdot ST_2 \dots ST_m}{s^p ST_a \cdot ST_b \dots ST_n}$$

$$= \frac{K_1 s^m}{s^n} = \frac{K_1}{s^{n-m}}$$

$$G(s) \cdot H(s) \bigg|_{s = \lim_{R \rightarrow \infty} R \cdot e^{j\theta}} = \frac{K_1}{\lim_{R \rightarrow \infty} (R e^{j\theta})^{n-m}}$$

$$= \underline{\underline{0 \cdot e^{-j\theta(n-m)}}}$$

Starting point

when $\theta = +\pi/2$

$$G(s)H(s) = 0 \cdot e^{-j\pi/2(n-m)}$$

End point

$$\theta = -\pi/2 \quad G(s) \cdot H(s) = 0 \cdot e^{+j\pi/2(n-m)}$$

Starting pt

$$-\pi/2(n-m) \rightarrow \pi/2(n-m) \text{ end pt}$$

$G(s)H(s)$ plane

Mapping of C_4

$$G(s) \cdot H(s) = \frac{K(1+ST_1)(1+ST_2) \dots}{s^P(1+ST_a)(1+ST_b) \dots}$$

$$S = \lim_{R \rightarrow \infty} R \cdot e^{j\theta} \quad \theta = -\frac{\pi}{2} \text{ to } +\frac{\pi}{2}$$

$$1+ST \approx 1$$

$$G(s) \cdot H(s) = \frac{K \cdot 1 \cdot 1 \dots}{s^P \cdot 1 \cdot 1} = \frac{K}{s^P}$$

$$S = \lim_{R \rightarrow \infty} R \cdot e^{j\theta}$$

$$G(s)H(s) = \frac{K}{\lim_{R \rightarrow \infty} (R e^{j\theta})^P} = \frac{K}{\lim_{R \rightarrow \infty} R^P \cdot e^{j\theta P}}$$

$$= \underline{\underline{\propto e^{-j\theta P}}}$$

when $\theta = -\pi/2$

$$G(s)H(s) = \infty e^{+j\pi/2 P}$$

Arc - radius ∞

$$+\frac{\pi}{2}P \text{ to } -\frac{\pi}{2}P$$

when $\theta = +\pi/2$

$$G(s)H(s) = \infty e^{-j\frac{\pi}{2}P}$$

p-no: of poles @ origin

Q) Draw the Nyquist plot for the s/m whose open loop transfer function is

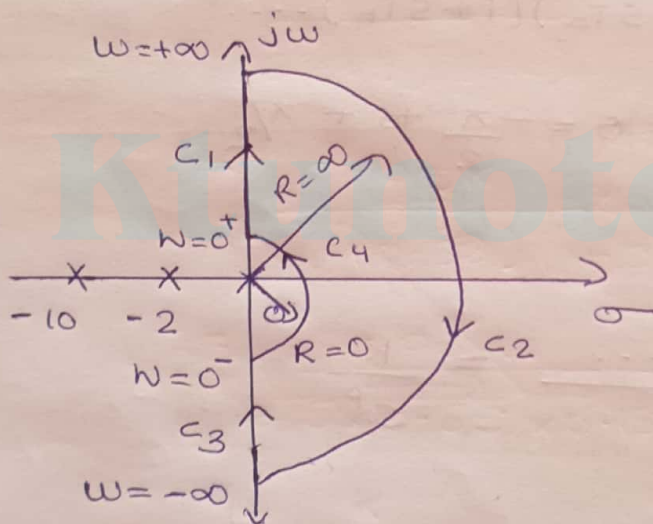
$$G(s)H(s) = \frac{K}{s(s+2)(s+10)}$$

Determine the range of K for which the closed loop s/m is stable

$$\begin{aligned} G(s)H(s) &= \frac{K}{s \times 2(1+0.5s) \times 10(1+0.1s)} \\ &= \frac{0.05K}{s(1+0.5s)(1+0.1s)} \end{aligned}$$

Select Nyquist contour :-

$$= \frac{0.05K}{-0.6\omega^2 + j\omega(1-0.05\omega^2)}$$

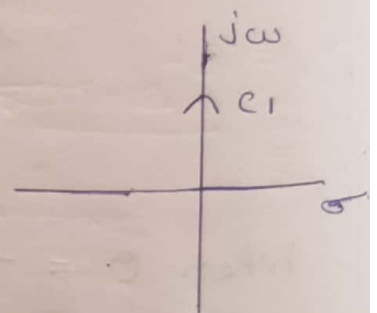


Mapping of section C1

$$G(s)H(s) = \frac{0.05K}{s(1+0.5s)(1+0.1s)}$$

$$s = j\omega$$

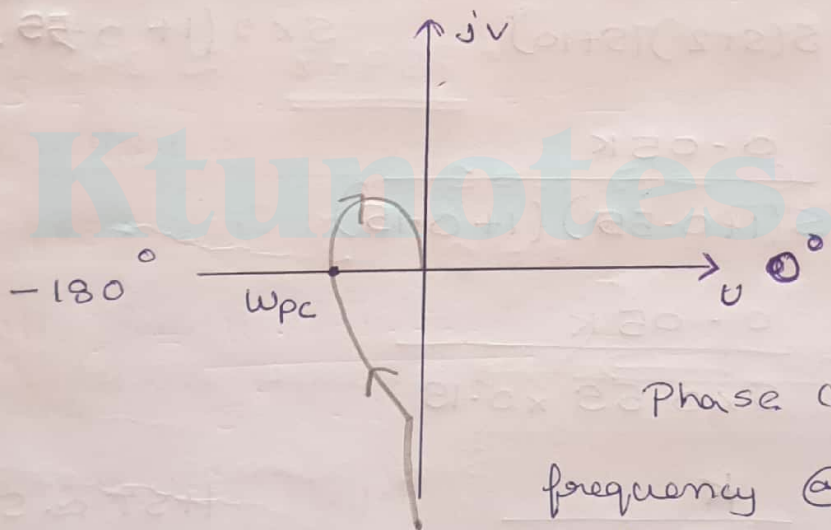
$$G(j\omega)H(j\omega) = \frac{0.05K}{j\omega(1+0.5j\omega)(1+0.1j\omega)}$$



$$|G(j\omega)H(j\omega)| = \frac{0.05K}{\omega \cdot \sqrt{1+(0.5\omega)^2} \sqrt{1+(0.1\omega)^2}}$$

$$\begin{aligned}\angle G(j\omega)H(j\omega) &= 0^\circ - 90^\circ - \tan^{-1}(0.5\omega) - \tan^{-1}(0.1\omega) \\ &= -90^\circ - \tan^{-1}(0.5\omega) - \tan^{-1}(0.1\omega)\end{aligned}$$

ω	$ G(j\omega)H(j\omega) $	$\angle G(j\omega)H(j\omega)$
$\omega = 0$	∞	-90°
1	0.04K	-122°
5	0.003K	-185°
∞	0	-270°



ω_{pc}
Phase cross over ω

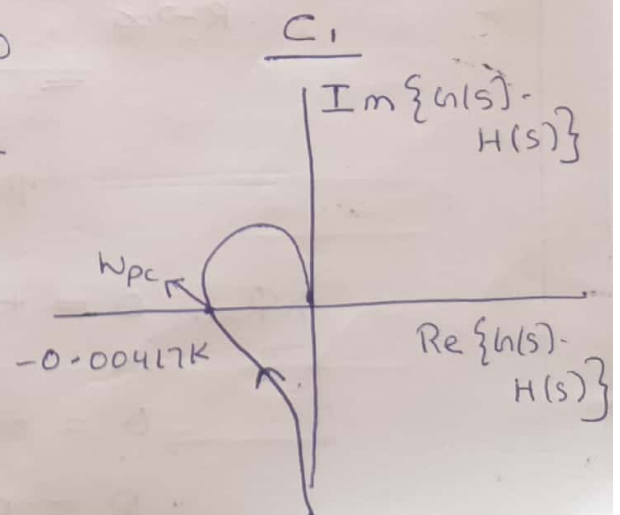
frequency @ which phase of s/m crosses ± 1

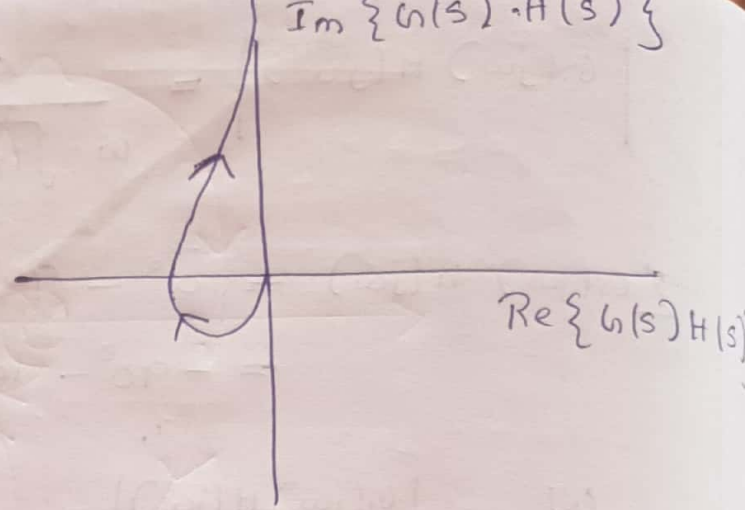
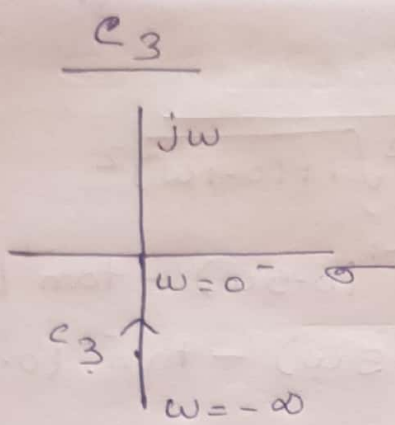
@ $\omega = \omega_{pc}$

$$\omega_{pc} (1 - 0.05\omega_{pc}^2) = 0$$

$$1 - 0.05\omega_{pc}^2 = 0$$

$$\omega_{pc} = \underline{\underline{4.47 \text{ rad/sec}}}$$





C₂

$$S = \lim_{R \rightarrow \infty} R e^{j\theta}$$

$$\theta = +\frac{\pi}{2} \text{ to } -\pi/2$$

$$G(s)H(s) = \frac{K}{s(s+2)(s+10)} = \frac{K}{s \times 2(1+0.5s)(1+0.5s)} \times 10$$

$$= \frac{0.05K}{s(1+0.5s)(1+0.1s)}$$

$$= \frac{0.05K}{s \times 0.5s \times 0.1s}$$

$$= \frac{K}{s^3}$$

$$1+sT \approx sT$$

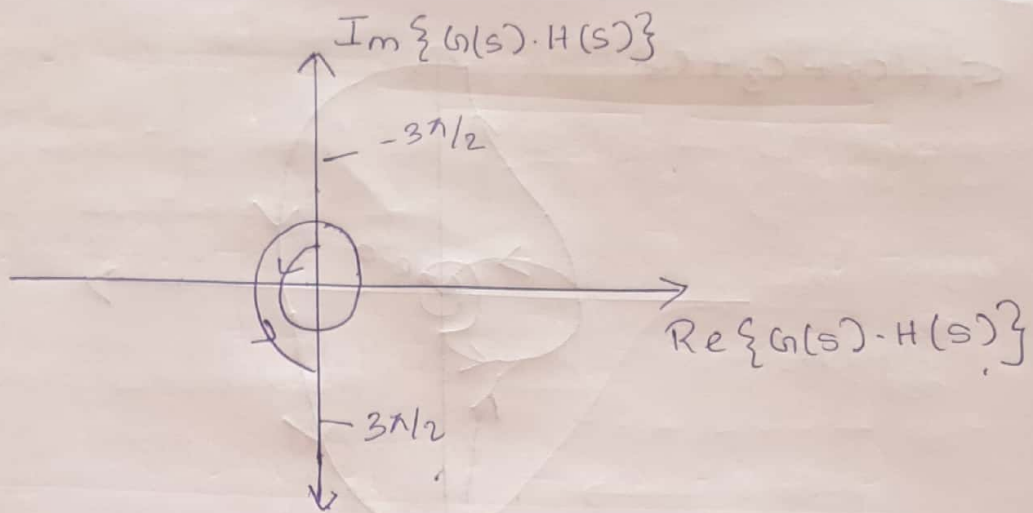
So $S = \lim_{R \rightarrow \infty} R e^{j\theta}$

$$= \frac{K}{\lim_{R \rightarrow \infty} R^3 e^{j3\theta}} = 0 \cdot e^{-j3\theta}$$

When $\theta = +\frac{\pi}{2} \rightarrow 0 \cdot e^{-j3\pi/2}$

$\theta = -\pi/2 \rightarrow 0 \cdot e^{j3\pi/2}$

$\theta = 0 \rightarrow 0 \cdot e^{-j0}$



mapping of C_4

$$s = \lim_{R \rightarrow 0} R e^{j\theta}$$

$$\theta = -\frac{\pi}{2} \text{ to } \frac{\pi}{2}$$

$$1 + ST \approx 1$$

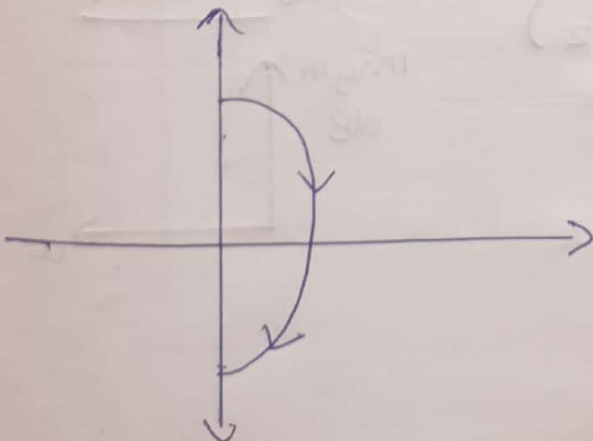
$$G(s) \cdot H(s) = \frac{K \times 0.05}{s(1)(1)}$$

$$\text{Sub } \lim_{R \rightarrow 0} R e^{j\theta} = \frac{0.05 K}{\lim_{R \rightarrow 0} R \cdot e^{j\theta}}$$

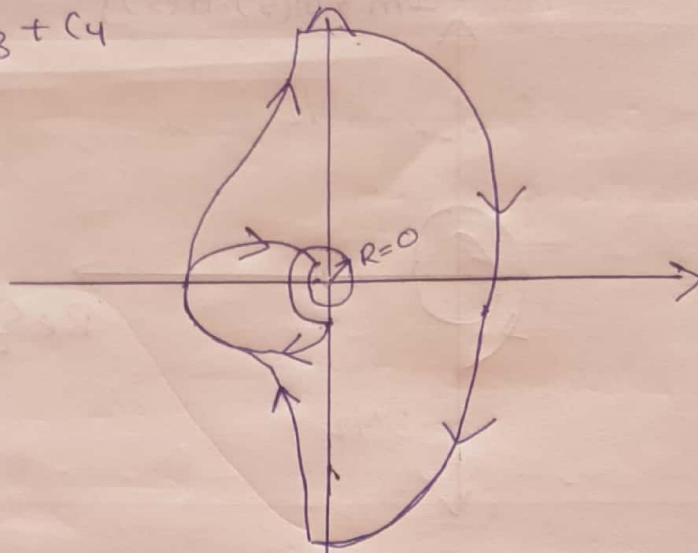
$$= \infty e^{-j\theta}$$

$$\text{When } \theta = -\pi/2 \quad \infty e^{j\pi/2}$$

$$\theta = +\pi/2 \quad \infty e^{-j\pi/2}$$



$$C_1 + C_2 + C_3 + C_4$$



For limiting value of K

$$\Rightarrow -0.00417K = -1$$

$$K = \frac{1}{0.00417}$$

$$= 239.8 \approx 240$$

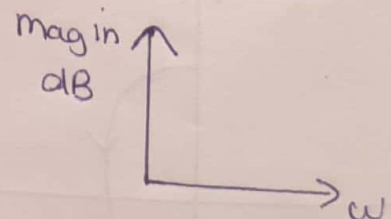
$$K < 240 \rightarrow \left. \begin{array}{l} N=0 \\ P=0 \end{array} \right\} \text{Stable s/m}$$

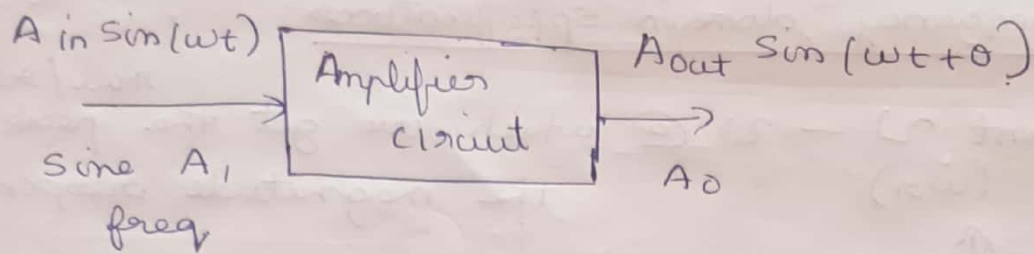
$K > 240 \rightarrow 2 \text{ clockwise encirclements}$
s/m always unstable

Bode plot - response of the s/m

$$G(s)H(s) = \frac{K(1+ST_1)}{S(1+ST_2)} \quad \text{--- ①} \quad \text{magnitude plot}$$

Phase plot





High ω 's can be plotted in semilog graph
 x axis - log scale
 y axis - ordinary

Sub $s = j\omega$ in (i)

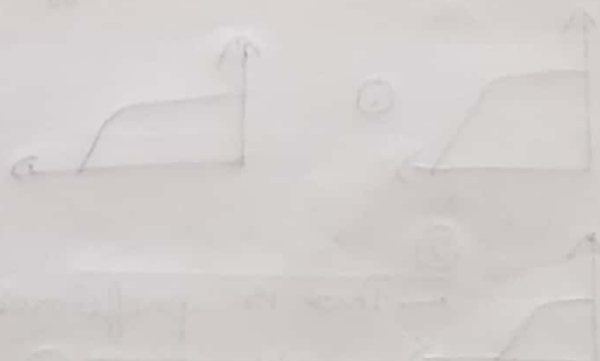
$$G(j\omega) H(j\omega) = \frac{K (1 + j\omega T_1)}{j\omega (1 + j\omega T_2)}$$

$$|G(j\omega) H(j\omega)| = \frac{K \sqrt{1 + (\omega T_1)^2}}{\omega \sqrt{1 + (\omega T_2)^2}}$$

$$|G(j\omega) H(j\omega)|_{\text{dB}} = 20 \log \left[\frac{K \sqrt{1 + (\omega T_1)^2}}{\omega \sqrt{1 + (\omega T_2)^2}} \right]$$

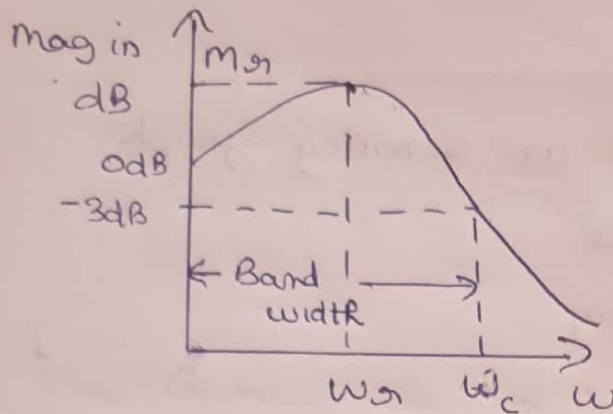
$$\begin{aligned} \angle G(j\omega) H(j\omega) &= \tan^{-1} \omega T_1 - \tan^{-1} \frac{\omega}{0} - \tan^{-1} \omega T_2 \\ &= \tan^{-1} \omega T_1 - 90 - \tan^{-1} \omega T_2 \end{aligned}$$

ω	ϕ
0	?
0.1	?
0.5	?
1	?



Frequency domain specifications

- 1) Resonant ω — ω @ which we get the ^{max/resonant} peak of the magnitude response

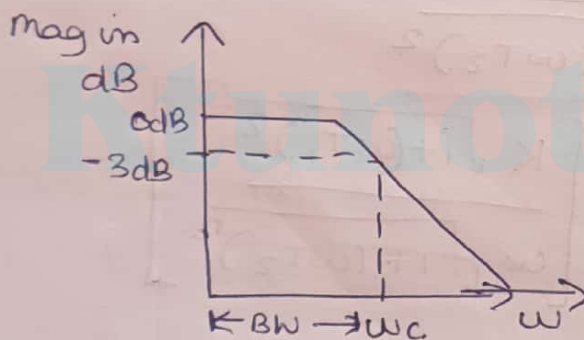


Magnitude plot.

Actual plot

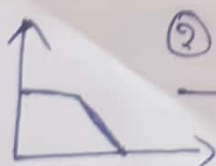
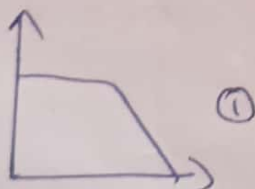
- 2) Resonant peak (M_m) — peak value of the magnitude response.

- 3) Bandwidth = $\omega_c - 0 = \omega_c$ Hz



Approximate plot.

- 4) Cut off ω — ω @ which you have gain of -3.
- 5) Cut off rate — Slope of magnitude response @ cut off ω .



— This is preferred
b/w ① & ②

6) gain margin

7) phase margin

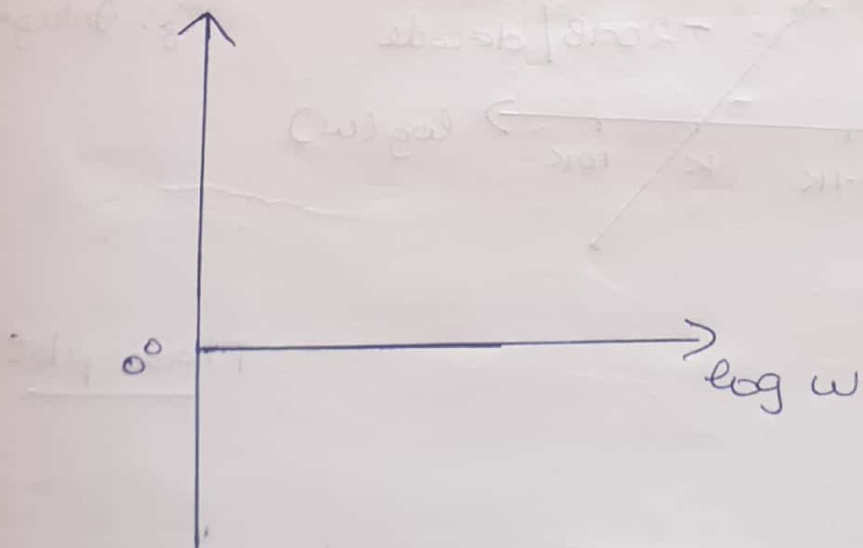
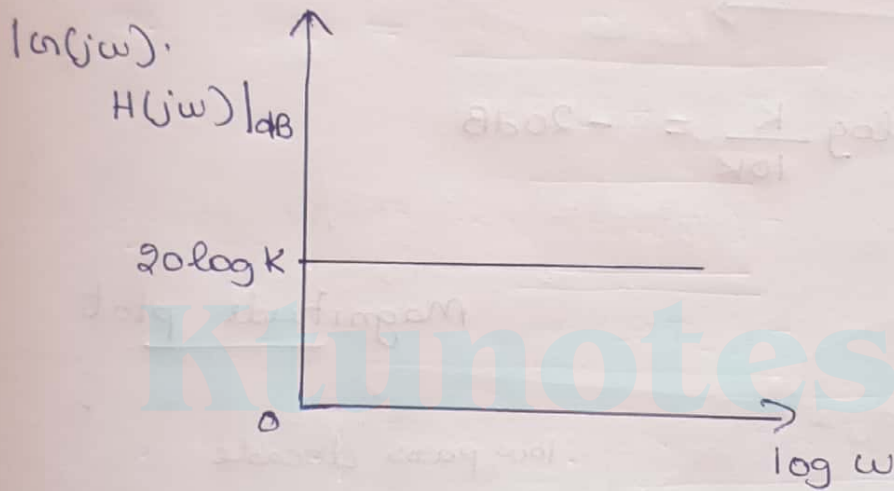
Q) $G(s)H(s) = K$

$$s = j\omega$$

$$G(j\omega)H(j\omega) = K$$

$$|G(j\omega)H(j\omega)|_{dB} = 20 \log K$$

$$\angle G(j\omega)H(j\omega) = 0^\circ$$



Q) $G(s)H(s) = \frac{K}{s}$ $\frac{1}{s}$ term - Integral factor

$$s = j\omega$$

$$G(j\omega)H(j\omega) = \frac{K}{j\omega}$$

$$|G(j\omega)H(j\omega)| = \frac{K}{\omega} \quad |G(j\omega)H(j\omega)|_{dB} = 20 \log \left(\frac{K}{\omega} \right)$$

$$\angle G(j\omega)H(j\omega) = -90^\circ$$

$$\omega = 0.1K$$

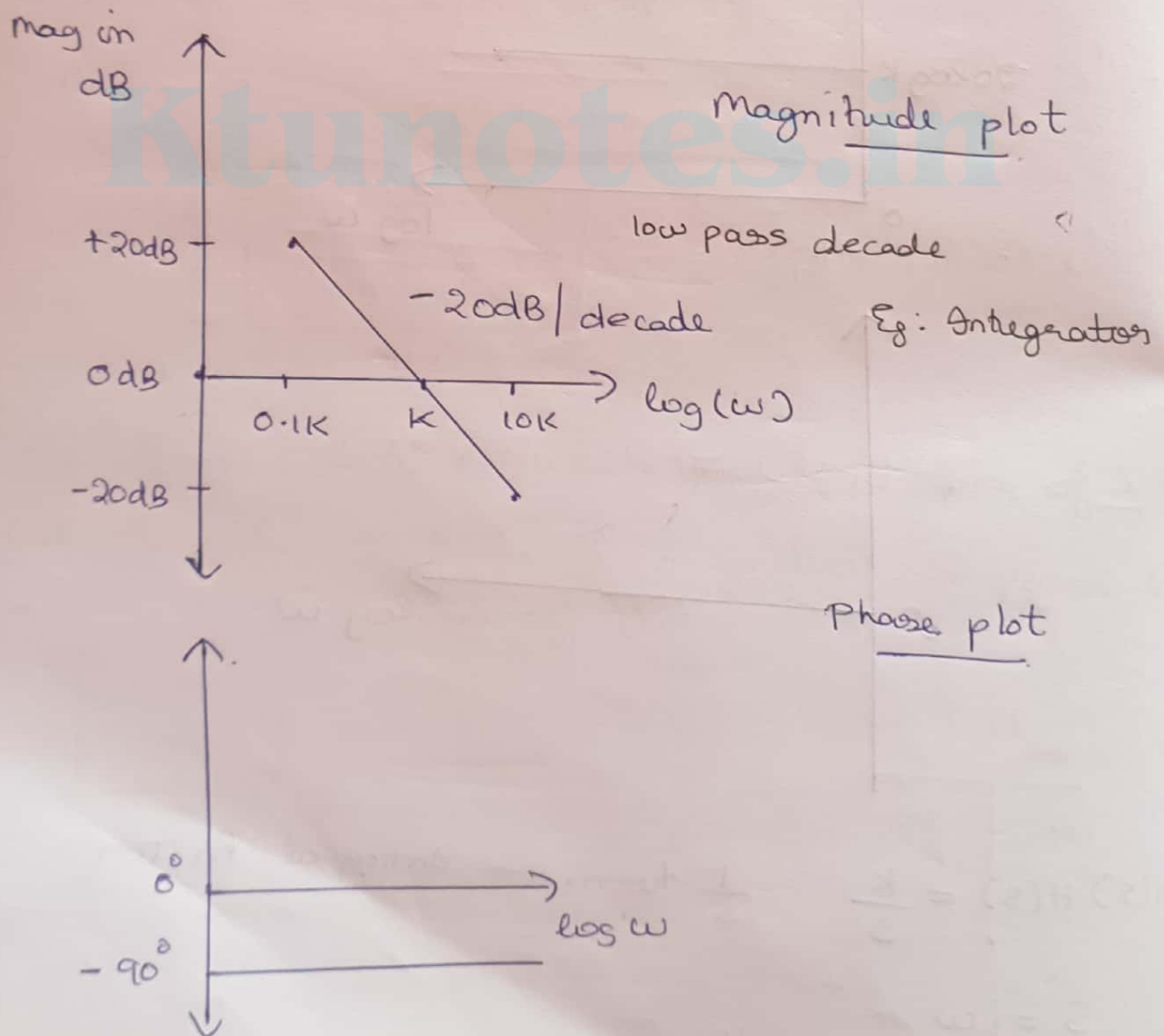
$$\text{mag dB} = 20 \log \frac{K}{0.1K} = +20 \text{ dB}$$

$$\omega = K$$

$$\text{mag dB} = 20 \log 1 = 0 \text{ dB}$$

$$\omega = 10K$$

$$\text{mag dB} = 20 \log \frac{K}{10K} = -20 \text{ dB}$$



$$3) G(s)H(s) = \frac{K}{s^n}$$

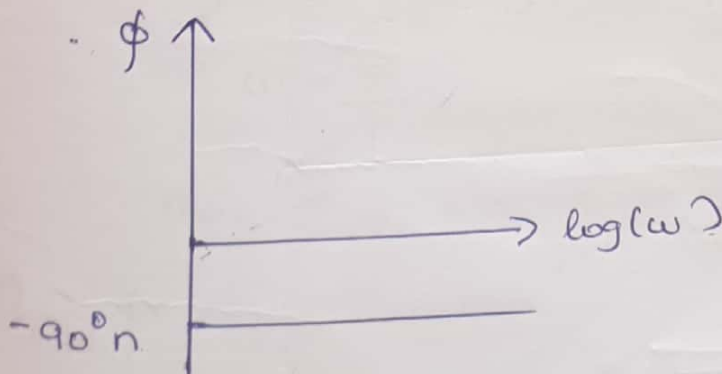
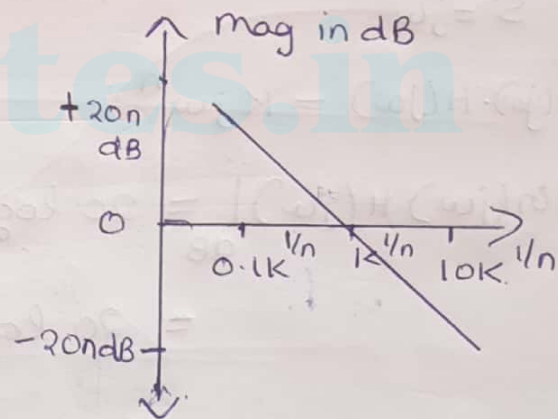
$$s = j\omega$$

$$|G(j\omega)H(j\omega)| = \frac{K}{(j\omega)^n} = \frac{K}{\omega^n}$$

$$\begin{aligned} |G(j\omega)H(j\omega)|_{dB} &= 20 \log \left(\frac{K}{\omega^n} \right) \\ &= 20 \log \left(\frac{K^{1/n}}{\omega} \right)^n \\ &= \underline{\underline{20n \log \frac{K^{1/n}}{\omega}}} \end{aligned}$$

$$\angle G(j\omega)H(j\omega) = -90^\circ \times n$$

ω	mag in dB
$0.1 K^{1/n}$	$+20n \text{ dB}$
$K^{1/n}$	0
$10 K^{1/n}$	$-20n \text{ dB}$



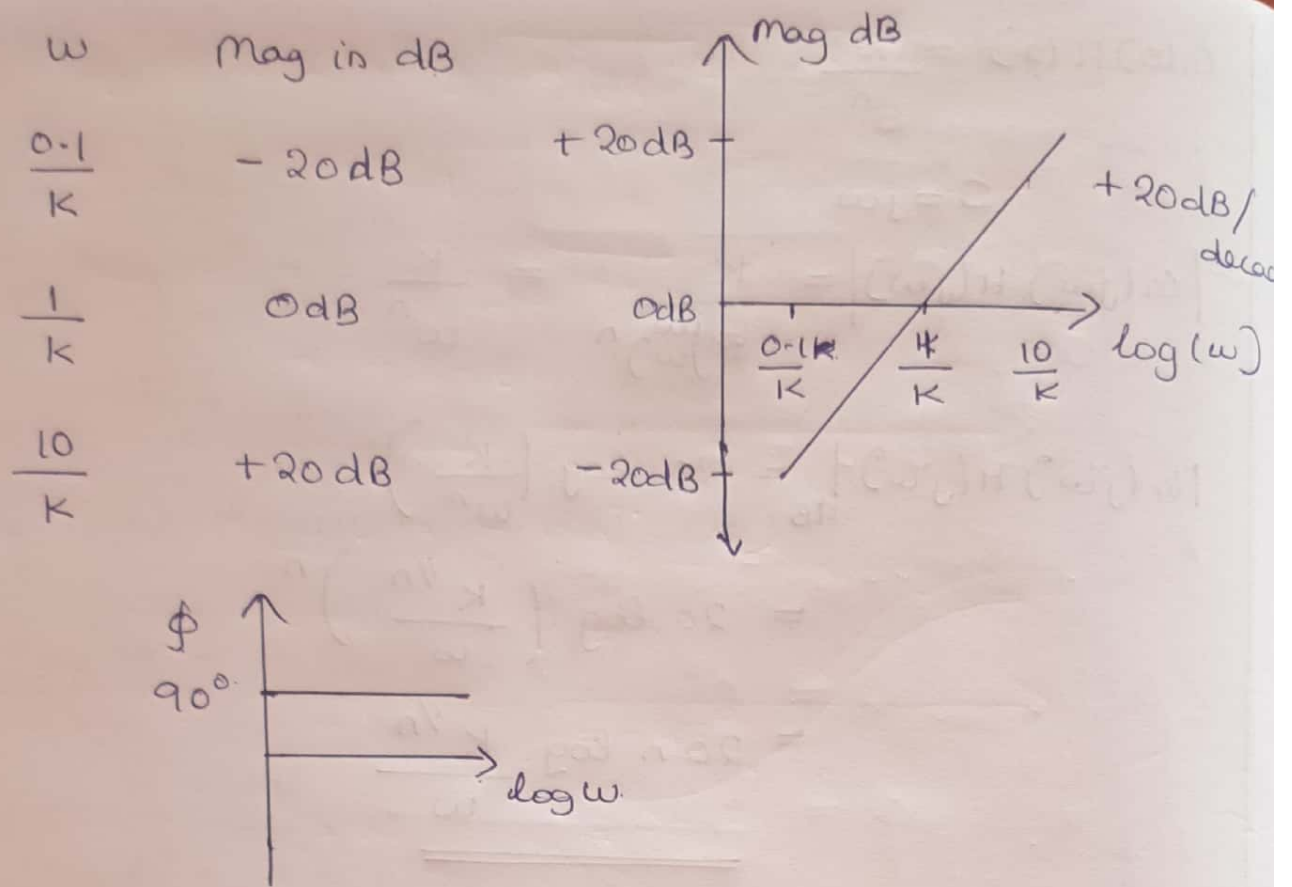
$$4) G(s)H(s) = Ks$$

$$s = j\omega$$

$$G(j\omega)H(j\omega) = K(j\omega)$$

$$|G(j\omega)H(j\omega)|_{dB} = 20 \log(K\omega)$$

$$\angle G(j\omega)H(j\omega) = 90^\circ$$



Q)

$$G(s) \cdot H(s) = Ks^n$$

$$s = j\omega$$

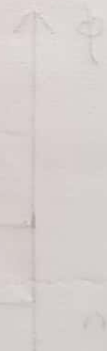
$$\text{Mag dB} = +20n \text{ dB/decade}$$

$$\text{Phase} = 90^\circ \cdot n$$

$$G(j\omega) \cdot H(j\omega) = K j\omega^n$$

$$|G(j\omega) H(j\omega)|_{\text{dB}} = 20 \log(K j\omega^n)$$

$$= 20 \log$$



$$G(s)H(s) = \frac{K}{1+ST}$$

$$S = j\omega$$

$$G(j\omega)H(j\omega) = \frac{K}{1+j\omega T}$$

$$|G(j\omega)H(j\omega)| = \frac{K}{\sqrt{1+(\omega T)^2}}$$

$$\begin{aligned}\angle G(j\omega)H(j\omega) &= 0^\circ - \tan^{-1}(\omega T) \\ &= -\tan^{-1}(\omega T)\end{aligned}$$

$$\omega T \ll 1$$

$$\text{Mag in dB} = 20 \log \frac{1}{\sqrt{1+(\omega T)^2}}$$

$$\omega T \ll 1$$

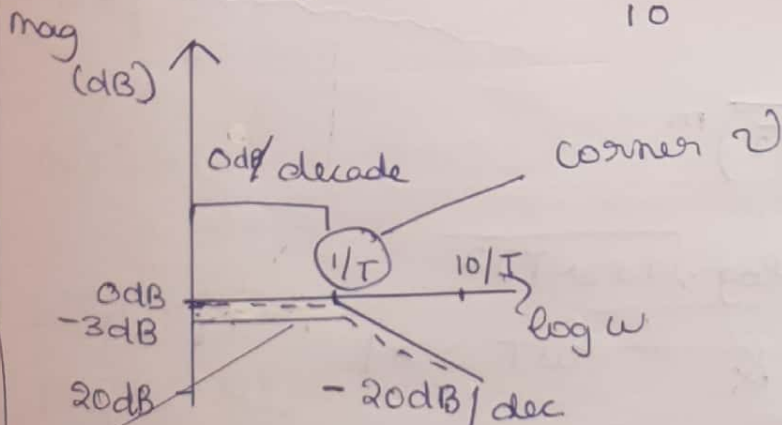
$$\text{mag (dB)} = 20 \log \frac{1}{\omega T}$$

$$\text{Let } \omega = \frac{1}{T}$$

$$\text{mag} = 20 \log 1 = \underline{\underline{0 \text{ dB}}}$$

$$\omega = \frac{10}{T}$$

$$\text{mag} = 20 \log \frac{1}{10} = \underline{\underline{-20 \text{ dB}}}$$



$$\text{mag dB} = -20 \log \sqrt{1 + (\omega T)^2}$$

$$\omega = 1/T$$

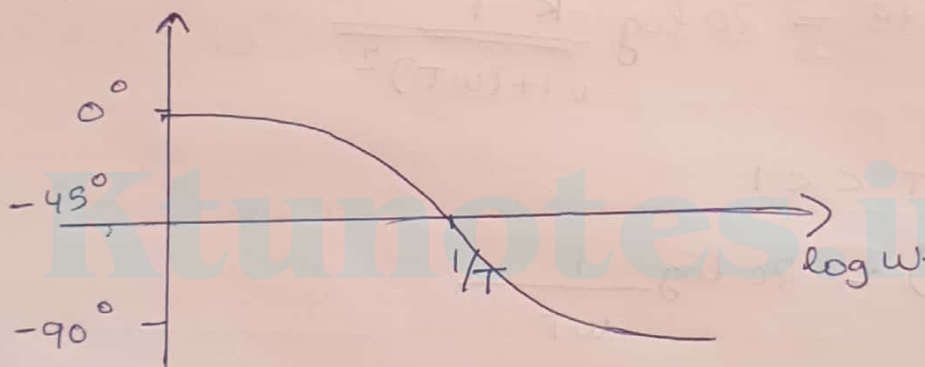
$$\begin{aligned} \text{mag dB} &= -20 \log \sqrt{1+1} \\ &= -3 \text{ dB} \end{aligned}$$

$$\angle G(j\omega) \cdot H(j\omega) = -\tan^{-1}(\omega T)$$

$$\omega = 0, \phi = 0^\circ$$

$$\omega = 1/T, \phi = -45^\circ$$

$$\omega = \infty, \phi = -90^\circ$$



$$Q) \quad G(s) \cdot H(s) = \frac{1}{(1+sT)^m}$$

$$s = j\omega$$

$$G(j\omega) H(j\omega) = \frac{1}{(1+j\omega T)^m}$$

$$|G(j\omega) H(j\omega)| = \frac{1}{(\sqrt{1+(\omega T)^2})^m}$$

$$\text{mag in dB} = -20m \log \sqrt{1+(\omega T)^2}$$

$$\omega T \ll 1 \quad \& \quad \omega T \gg 1$$

$$\text{mag in dB} = \underline{\underline{0 \text{ dB}}} \quad - \omega T \ll 1$$

$$\omega T \gg 1$$

$$\text{mag in dB} = -20m \log \omega T$$

$$@ \omega = \frac{1}{T}$$

$$\text{mag dB} = \underline{\underline{0 \text{ dB}}}$$

$$@ \omega = \frac{10}{T}$$

$$\text{mag dB} = \underline{\underline{-20 m \text{ dB}}}$$

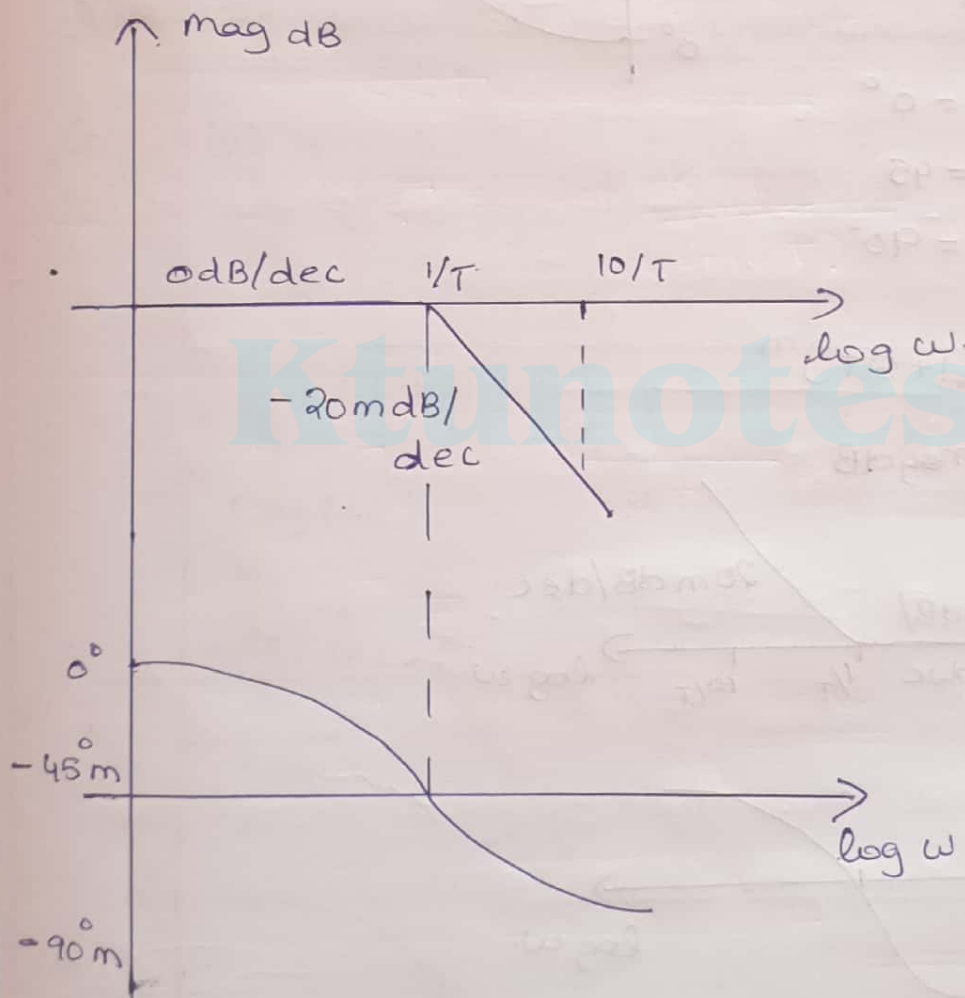
$$\angle G(j\omega) \cdot H(j\omega) =$$

$$-m \tan^{-1}(\omega T)$$

$$\omega = 0, \phi = 0^\circ$$

$$\omega = 1/T, \phi = -45^\circ \times m$$

$$\omega = \infty, \phi = -90^\circ \times m$$



$$G(s)H(s) = 1 + sT$$

$$s = j\omega$$

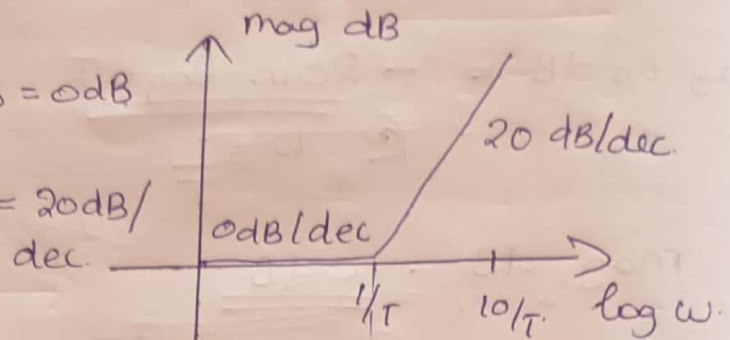
$$G(j\omega)H(j\omega) = 1 + j\omega T$$

$$|G(j\omega)H(j\omega)| = \sqrt{1 + (\omega T)^2}$$

$$\text{Mag in dB} = 20 \log \sqrt{1 + (\omega T)^2}$$

$$\omega = \frac{1}{T}, \text{ mag dB} = 0 \text{ dB}$$

$$\omega = \frac{10}{T}, \text{ mag dB} = 20 \text{ dB/dec}$$

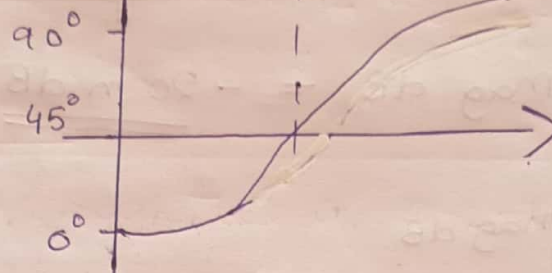


$$\phi = \tan^{-1} \omega T$$

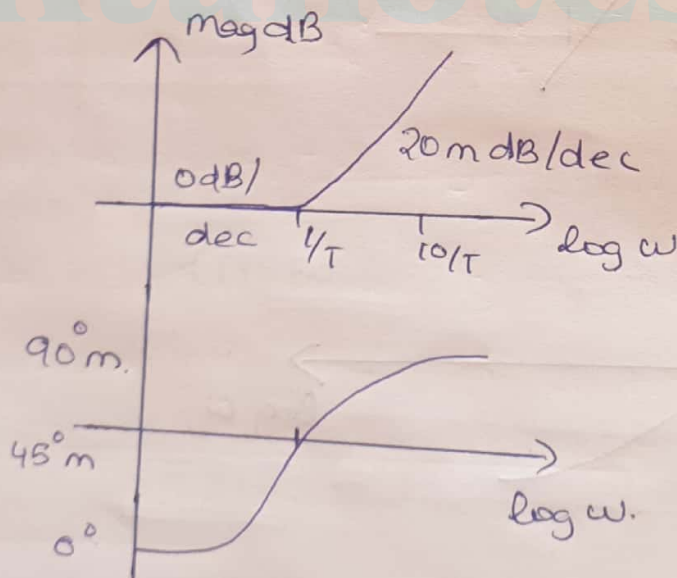
$$\omega = 0, \phi = 0^\circ$$

$$\omega = 1/T, \phi = 45^\circ$$

$$\omega = \infty, \phi = 90^\circ$$



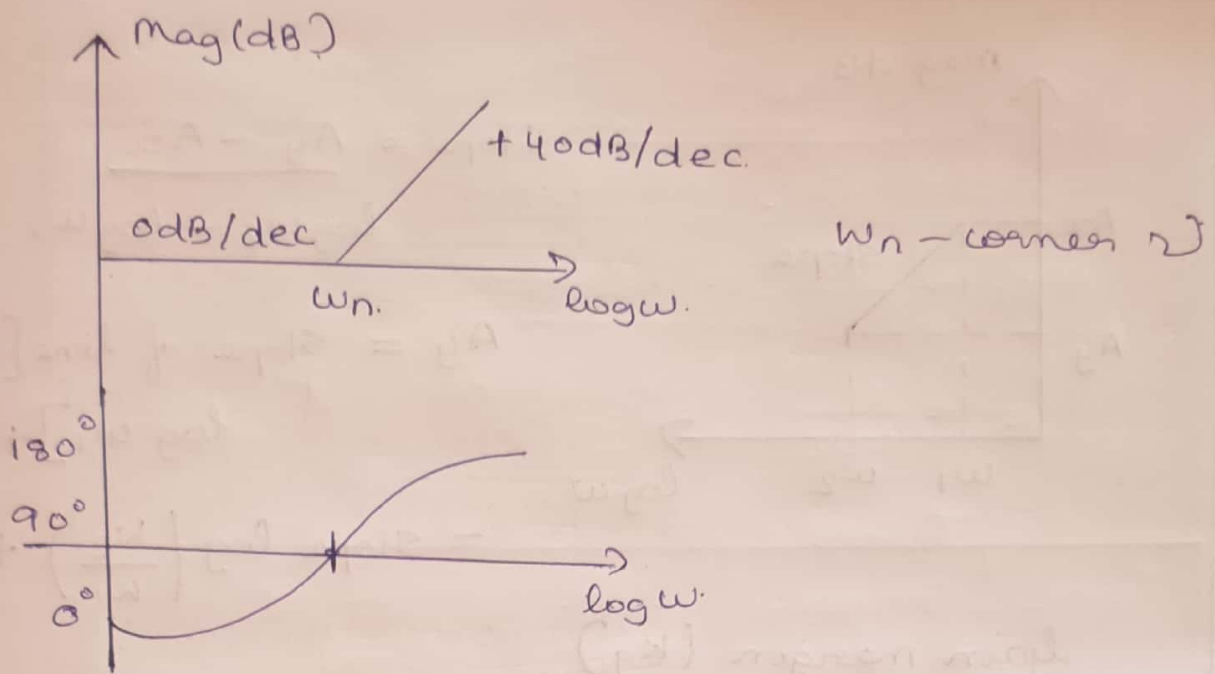
$$\phi) G(s)H(s) = (1 + sT)^m$$



Quadratic factor in the numerator

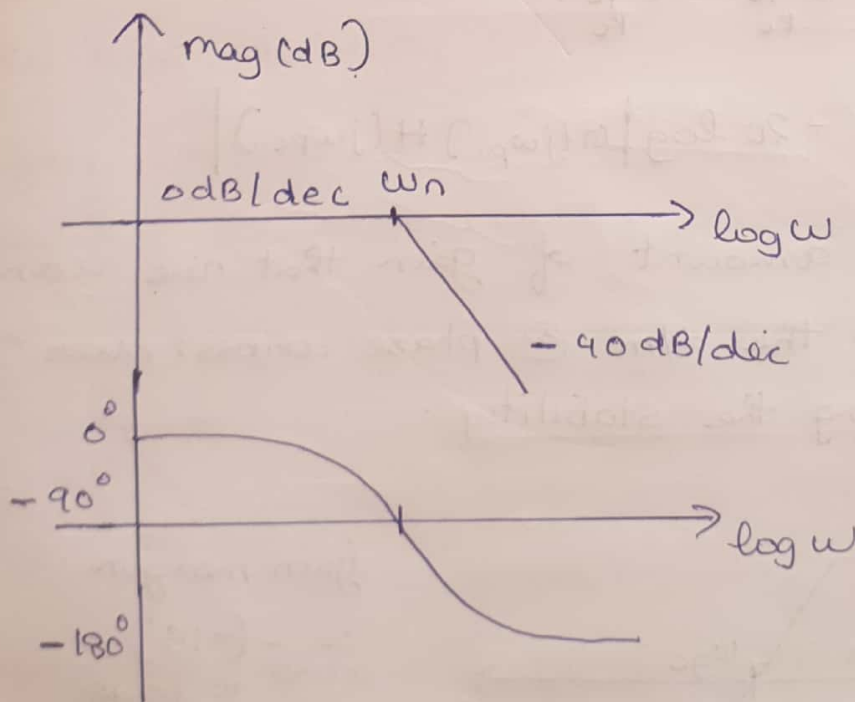
$$G(s) \cdot H(s) = \frac{s^2 + 2\zeta \omega_n s + \omega_n^2}{\omega_n^2}$$

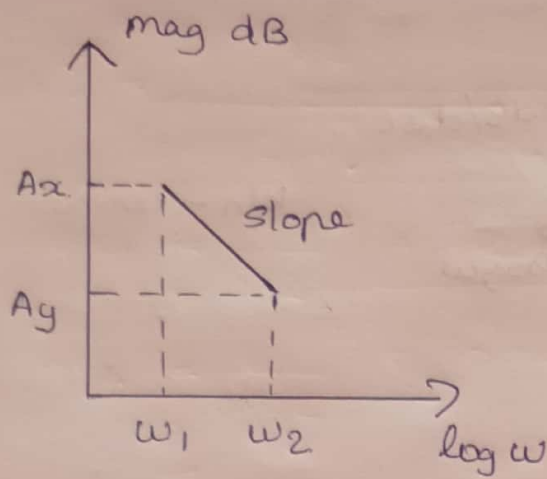
$$= 1 + 2\zeta \left(\frac{s}{\omega_n}\right) + \left(\frac{s}{\omega_n}\right)^2$$



Quadratic factor in the denominator

$$\begin{aligned}
 G(s)H(s) &= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \\
 &= \frac{1}{1 + 2\zeta\left(\frac{s}{\omega_n}\right) + \left(\frac{s}{\omega_n}\right)^2}
 \end{aligned}$$





$$\text{Slope} = \frac{A_y - A_x}{\log w_2 - \log w_1}$$

$$A_y = \text{Slope of line} [\log w_2 - \log w_1] + A_x$$

$$= \text{Slope} \log \left(\frac{w_2}{w_1} \right) + A_x$$

Gain margin (K_g)

* $K_g \rightarrow$ Reciprocal of magnitude of O.L.T.F @

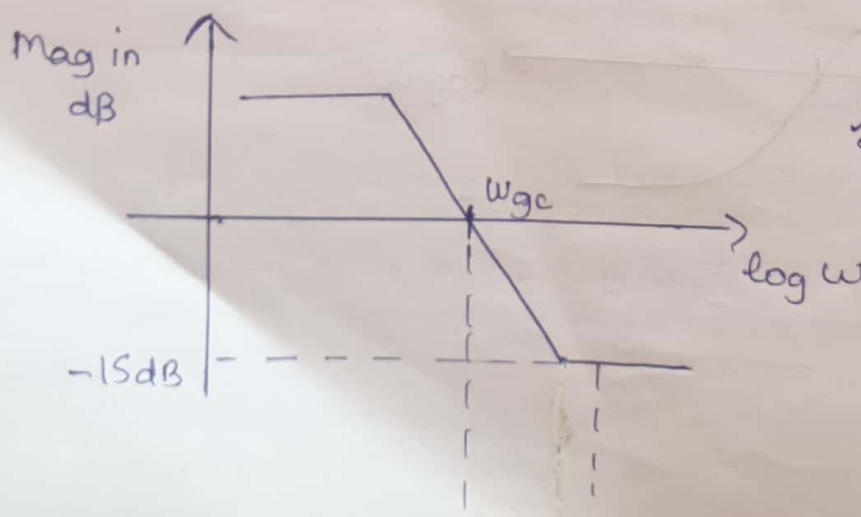
* Phase cross over frequency.

Phase cross over $\omega - \omega$ where phase of S/m crosses $\pm 180^\circ$

$$K_g = \frac{1}{|G(j\omega_{pc}) H(j\omega_{pc})|}$$

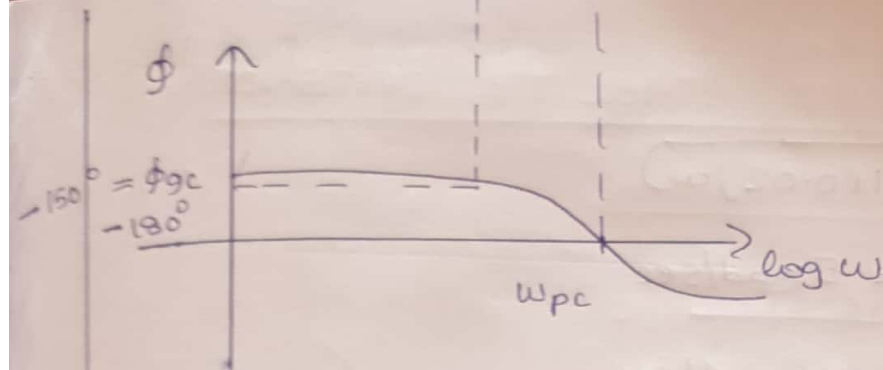
$$\text{G.M. in dB} = -20 \log |G(j\omega_{pc}) H(j\omega_{pc})|$$

* $K_g \rightarrow$ additional amount of gain that we can introduce into the S/m @ phase cross over ω without affecting the stability.



$$\text{Gain margin} = -(-15)$$

$$= \underline{\underline{15 \text{ dB}}}$$



$$C.L.T.F = \frac{G(s)}{1 + G(s)H(s)}$$

$$1 + G(s)H(s) = 0$$

$$G(s)H(s) = -1$$

$$|G(s)H(s)| = 1 = 0 \text{ dB}$$

$$\angle G(s)H(s) = -180^\circ$$

Phase margin (ν)

Additional phase lag that can be introduced into the s/m @ gain cross over ν without affecting the s/m stability.

* gain cross over ν (ω_{gc}) — ν @ which the gain of the open loop T.F becomes 0 dB

$$PM(\nu) = 180^\circ + \phi_{gc}$$

$$\phi_{gc} = -150^\circ$$

$$\underline{\underline{PM(\nu) = 30^\circ}}$$

* It can be — we also

Q) Sketch the bode plot for the following T.F & determine the gain of the s/m K such that the gain cross over ν is 5 rad/sec? $G(s) = \frac{Ks^2}{(1+0.2s)(1+0.02s)}$

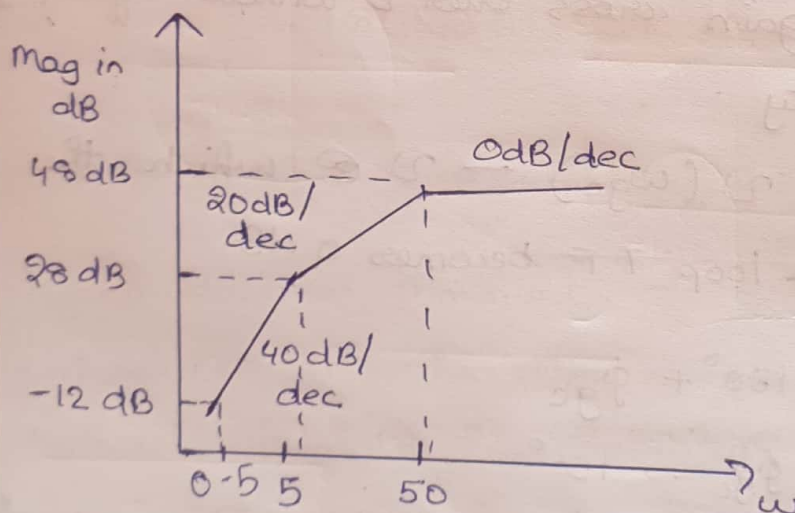
$$S = j\omega$$

$$G(j\omega) = \frac{(j\omega)^2}{(1 + 0.2j\omega)(1 + 0.02j\omega)}$$

$$\omega_{c1} = \frac{1}{0.2} = 5 \text{ rad/sec}$$

$$\omega_{c2} = \frac{1}{0.02} = 50 \text{ rad/sec}$$

Terms	Corners ω rad/sec	Slope dB/dec	change in slope dB/dec
$(j\omega)^2$	—	40 dB/dec	—
$\frac{1}{1 + 0.2j\omega}$	5 rad/sec	-20 dB/dec	40 - 20 = 20 dB/dec
$\frac{1}{1 + 0.02j\omega}$	50 rad/sec	-20 dB/dec	20 - 20 = 0 dB



ω_L - lower of
lower cut off ω

$$\omega_L, \omega_{c1}, \omega_{c2}, \omega_H$$

$$\text{Let } \omega_L = 0.5 \text{ rad/sec}$$

$$\begin{aligned} \text{mag dB} &= 20 \log \omega_L^2 \\ &= -12 \text{ dB} \end{aligned}$$

$$@ \omega = \omega_{c1} = 5 \text{ rad/sec}$$

$$\text{Mag (dB)} = (\text{slope of line from } 0.5 \text{ rad/sec to } 5 \text{ rad/sec} \times \log \left(\frac{\omega_{c1}}{\omega_{c2}} \right) + \text{mag (dB)} @ \omega = \omega_{c1})$$

$$= 40 \times \log \left(\frac{5}{0.5} \right) - 12 \text{ dB}$$

$$= \underline{\underline{28 \text{ dB}}}$$

$$@ \omega = \omega_{c2}$$

$$\text{mag (dB)} = \text{slope of line from } \omega_{c1} \text{ to } \omega_{c2} \log \left(\frac{\omega_{c2}}{\omega_{c1}} \right) + \text{mag (dB)} @ \omega = \omega_{c1}$$

$$= 20 \times \log \left(\frac{50}{0.5} \right) + 28 \text{ dB}$$

$$= \underline{\underline{48 \text{ dB}}}$$

$$\text{Let } \omega_h = 100 \text{ rad/sec, magnitude} = 48 \text{ dB}$$

Phase plot

$$G(j\omega) = \frac{(j\omega)^2}{(1+0.2j\omega)(1+0.02j\omega)}$$

$$\angle G(j\omega) = 180^\circ - \tan^{-1}(0.2\omega) - \tan^{-1}(0.02\omega)$$

ω	$\angle G(j\omega)$	gain (cross over ω)
0.1	178.739	$20 \log K = -28 \text{ dB}$ $K = \underline{\underline{0.0398}}$ we shall add 28dB to get 0dB @ 5 rad/sec 28dB 26dB - 28dB
0.5	173.716	
1	167.544	
5	129.289	
10	105.255	
20	82.23	@ 15 rad, $20 \log K = -38$ $K =$
50	50.71	
100	29.427	

Here $\omega_{gc} < \omega_{pc}$ — S/m stable

$$\text{Gain margin} = -20 \log d$$

$$= 20 \text{ dB}$$

$$\text{Phase margin} = 40 \text{ dB}$$

} +ve.

$\Rightarrow$ S/m stable

Q) Sketch bode plot & determine gain margin & phase margin?

$$G(s) = \frac{75(1+0.2s)}{S(s^2+16s+100)}$$

$$s = j\omega$$

$$G(j\omega) = 75 \left(\frac{1+0.2j\omega}{j\omega \cdot 100(1+0.16j\omega+0.01\omega^2)} \right)$$

$$G(s) = \frac{75(1+0.2s)}{S \cdot 100(1+0.16s+0.01s^2)}$$

$$s = j\omega$$

$$G(j\omega) = \frac{75(1+0.2j\omega)}{j\omega \cdot 100(1+0.16j\omega+0.01\omega^2)}$$

$$j\omega \cdot 100(1+0.16j\omega+0.01\omega^2)$$

$$\omega_{c1} = \frac{1}{T_1} = \frac{1}{0.2} = 5 \text{ rad/sec}$$

$$\omega_{c2} = \frac{1}{T_2} = \frac{1}{0.01} = 10 \text{ rad/sec}$$

$$\omega_n = 10 \text{ rad/sec}$$

Terms	Corner freq	Slope	Change in Slope
$\frac{0.75}{j\omega}$	—	-20dB/dec	
$(1+0.2j\omega)$	5 rad/sec	+20dB/dec	0dB/dec
$\frac{1}{(1-0.01\omega^2+0.16j\omega)}$	10 rad/sec	-40dB/dec	-40dB/dec

Let;

$$\omega_L = 0.1 \text{ rad/sec}$$

$$\omega_H = 30 \text{ rad/sec}$$

$$\omega = \omega_c = 0.1 \text{ rad/sec}$$

$$\begin{aligned} \text{mag in dB} &= 20 \log \left(\frac{0.75}{j\omega} \right) \\ &= \underline{\underline{+18 \text{ dB}}} \end{aligned}$$

$$\omega = \omega_{c1} = 5 \text{ rad/sec}$$

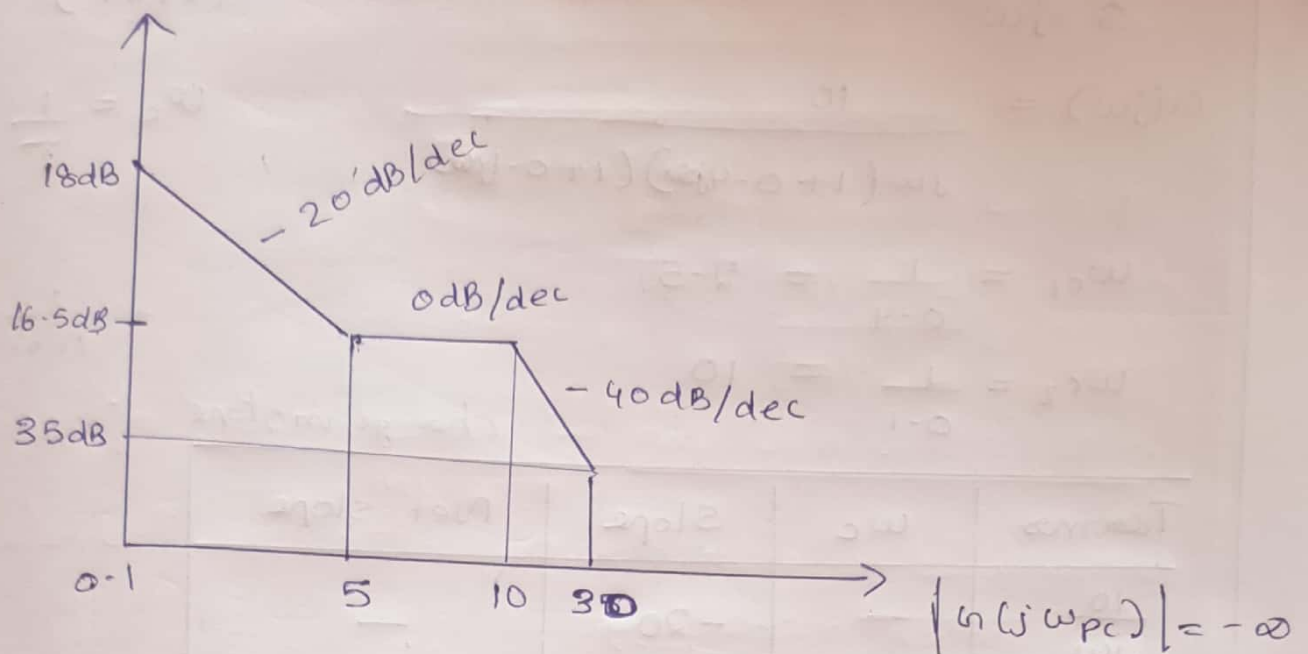
$$\begin{aligned} \text{mag in dB} &= -20 \times \log \left(\frac{5}{0.1} \right) + 18 \\ &= -15.9 \text{ dB} \\ &= \underline{\underline{-16 \text{ dB}}} \end{aligned}$$

$$@ \omega = \omega_R = 30 \text{ rad/sec}$$

$$\text{mag dB} = \underline{\underline{-35 \text{ dB}}}$$

$$\phi = \tan^{-1}(0.2\omega) - 90^\circ - \tan^{-1} \left(\frac{0.16\omega}{1 - 0.01\omega^2} \right)$$

ω	$\tan^{-1} 0.2\omega$	$\tan^{-1} \frac{0.16\omega}{1 - 0.01\omega^2}$	ϕ
0.1	1.1	0.9	-89.8
0.5	5.7	4.59	-88.8
1	11.3	9.1	-87.8
2	21.80	18.	-86.2
5	45.	46.8	-91.8
10	63.43	65.6 90°	-116.57
20	75.96	76 - 46.8° + 180°	-147
30	80.53	-30.9° + 180°	139.63
50	84.28	-18.4° + 180°	-167



$$\text{Gain margin} = -(-\infty) = \underline{\underline{+\infty}}$$

Phase margin

$$\omega_{gc} = 0.8 \text{ rad/sec}$$

$$\phi_{gc} = -88^\circ$$

$$\begin{aligned} \text{Phase margin} &= 180^\circ + \phi_{gc} \\ &= 180 - 88 \\ &= \underline{\underline{92^\circ}} \end{aligned}$$

$$G(s) = \frac{250}{s(s+2.5)(s+10)}$$

Find gain & phase cross
over ω ? margin

$$G(s) = \frac{250}{2.5 s \left(\frac{s}{2.5} + 1 \right) \left(\frac{s}{10} + \frac{10}{2.5} \right) 10}$$

$$\begin{aligned} G(s) &= \frac{250}{2.5 s (1+0.4s) \times 10 (0.1+0.1s)} \\ &= \frac{1.0}{s(1+0.4s)(1+0.1s)} \end{aligned}$$

$$S = j\omega$$

$$(1+sT)$$

$$G(j\omega) = \frac{10}{j\omega(1+0.4j\omega)(1+0.1j\omega)}$$

$$\omega_c = \frac{1}{T}$$

$$\omega_{c1} = \frac{1}{0.4} = 2.5$$

$$\omega_{c2} = \frac{1}{0.1} = 10$$

$$\frac{1}{1+sT}$$

change in slope

Terms	ω_c	Slope	Net slope
$\frac{10}{j\omega}$	—	-20	—
$\frac{1}{(1+0.4j\omega)}$	2.5	-20	-20-20 = -40 dB/dec
$\frac{1}{(1+0.1j\omega)}$	10	-20	-40-20 = -60 dB/dec

$$\omega_L, \omega_{c1}, \omega_{c2}, \omega_K$$

$$\omega_L = 0.1 \Rightarrow \text{mag} = 20 \log \frac{10}{0.1} = 40 \text{ dB}$$

$$\omega_{c1} = 2.5$$

$$-20 = \frac{40 - \alpha}{\log 0.1 - \log 2.5}$$

$$27.95 = 40 - \alpha$$

$$\alpha = 12 \text{ dB}$$

$$\omega_{c2} = 10$$

$$-40 = \frac{12 - \alpha}{\log 2.5 - \log 10}$$

$$\alpha = 12 - 24 = -12 \text{ dB}$$

$$\omega_K = 20$$

$$-60 = \frac{-12 - \alpha}{\log 10 - \log 20}$$

$$\alpha = -30.06 \text{ dB}$$

$$\phi = -90^\circ - \tan^{-1} 0.4\omega - \tan^{-1} 0.1\omega$$

ω	ϕ
0.1	-92.86°
0.5	-104.17°
1	-117.51°
5	-180°
10	-210.96°
15	-226.84°
20	-236.3°
30	-246.80°

$$\omega_{gc} = 5 \text{ rad/sec}$$

$$\omega_{pc} = 5 \text{ rad/sec}$$

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